



Bankruptcy prediction using the text-based communicative value of earnings call transcripts

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Abstract

We examine whether incorporating the text-based communicative value (*TCV*) of earnings call transcripts improves the effectiveness of bankruptcy prediction models within machine learning frameworks, utilizing U.S. firm data from 2005 to 2020. We find that the inclusion of earnings call transcripts *TCV* variables significantly improves the overall bankruptcy prediction effectiveness in addition to Barboza's et al. (Expert Syst Appl 83:405-417, 2017) financial variables. Notably, the incremental contribution of earnings call transcripts *TCV* variables is more pronounced in the future longer-term bankruptcy predictions, aligning with the forward-looking nature of earnings call transcripts and complementing the findings of Chen et al. (Expert Syst Appl 233:120714, 2023). Furthermore, the feature engineering results indicate that the improvement is mainly driven by positive tone and uncertainty tone variables, which more directly capture key determinants in structural-form credit risk models (e.g., asset value, volatility, and incomplete information). These tone variables signal a firm's prospective financial condition and future asset value distribution, thereby reflecting bankruptcy risk. Consistent with theoretical expectations, positive tone is negatively related to bankruptcy risk while uncertainty tone has the opposite effect. Finally, these results remain robust when annual report *TCV* variables are included as additional benchmark model input variables.

Keywords Earnings call transcripts · Text-based communicative value · Bankruptcy prediction · Machine learning · Bankruptcy risk term structure

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1 Introduction

Global financial markets have faced a series of severe challenges in recent years. From 2007 to 2009, the U.S. subprime mortgage crisis triggered a wave of corporate bankruptcies, deeply impacting global markets. In 2019, the U.S.-China trade war increased tariffs, disrupted supply chains, and led to price surges, causing widespread bankruptcies. The 2020 COVID-19 pandemic added further strain, with lockdowns triggering high unemployment and especially harming the service sector. These crises have led creditor banks to adopt overly conservative lending practices, reducing loan losses but limiting efficient capital use. This restricts credit access for otherwise strong companies, creating liquidity risks and intensifying supply chain contagion. Thus, accurately predicting corporate bankruptcy is now essential to mitigate credit losses, improve capital efficiency, and curb supply chain risks.

Bankruptcy prediction models have evolved from traditional statistical approaches (e.g., Beaver 1966; Altman 1968; Ohlson 1980) to machine learning models (e.g., Barboza et al. 2017; Mai et al. 2019; Kim and Yoon 2021; Chen et al. 2023, 2025a, 2025b), the latter now dominating recent advancements. Early studies primarily relied on corporate financial accounting data (Beaver 1966; Altman 1968). More recent research incorporates non-financial factors, such as corporate governance (Liang et al. 2017), textual analysis of annual reports (Mai et al. 2019; Chen et al. 2023, 2025a), and social network attributes of managers (Chen et al. 2024). Specifically, Mai et al. (2019) and Kim and Yoon (2021) identify word frequency in the Management Discussion and Analysis (hereafter denoted as MD&A) section of annual reports as a novel input variable for bankruptcy prediction models. Meanwhile, Chen et al. (2023) introduce the concept of communicative value in financial reports (Seebeck and Kaya 2023), proposing the text-based communicative value (TCV) of annual reports as an advanced natural language processing (hereafter denoted as NLP) feature to improve the accuracy of corporate bankruptcy prediction models. In addition, Chen et al. (2025a) also further employ the uncertainty of annual report *TCV* as another engineered NLP features. However, most of the existing studies focus on the textual information of annual reports, while the textual information of earnings call transcripts has rarely been mentioned and discussed. Unlike annual reports that only provide textual information about past corporate operation results, earnings conference calls can offer more timely, relevant, and forward-looking information about corporate activities. Moreover, the question-and-answer (namely discussion) sessions during earnings conference calls can reflect the difference in perceptions of a firm's asset value distribution between internal stakeholders (executive teams) and external stakeholders (analysts). Accordingly, the textual features found in earnings call transcripts hold a distinct communicative value compared to the content present in annual reports. This study follows Chen et al. (2023) to employ the readability and tones variables as the proxies of earnings call transcripts *TCV*.

Furthermore, because earnings call transcripts *TCV* captures management's forward-looking sentiment and real-time confidence, it conveys value-relevant forward-looking information through readability and tones. Accordingly, earnings call transcripts *TCV* may affect a firm's future asset value distribution and its incomplete information level. Meanwhile, based on the theoretical framework of structural-form credit models (e.g. Merton 1974; Duffie and Lando 2001), asset value, asset value volatility, the default threshold, and

incomplete information are four crucial factors influencing corporate credit risk. Therefore, using earnings call transcripts *TCV* variables for bankruptcy prediction is theoretically grounded. In contrast to previous literature utilizing machine learning models for bankruptcy prediction, this study aims to compare the predictive effectiveness of machine learning models by introducing earnings call transcripts *TCV* variables, in addition to traditional financial accounting variables (Barboza et al. 2017) and annual report *TCV* features (Chen et al. 2023). Therefore, this study explores whether earnings call transcripts *TCV* variables can help the bankruptcy prediction model capture more credit risk-related information, namely improve the model effectiveness of bankruptcy prediction.

Based on the above discussions, this study proposes the new engineered NLP measures of earnings call transcripts, namely earnings call transcripts *TCV* variables, as the proxies of value-relevant forward-looking signals and further incorporates them as additional input variables into the bankruptcy prediction model. Following the approaches of Seebeck and Kaya (2023) and Chen et al. (2023), this study uses readability and tones as the main proxies of earnings call transcripts *TCV*. Subsequently, following the machine learning model settings outlined by Barboza et al. (2017) and Chen et al. (2023), this study employs the 11 financial feature variables mentioned by Barboza et al. (2017) as benchmark model input variables. Furthermore, under the setting of the aforementioned 11 financial variables, the study additionally introduces earnings call transcripts *TCV* variables to implement machine learning model and investigates whether these earnings call transcripts *TCV* variables can significantly enhance the effectiveness of the bankruptcy prediction model. In addition, following Chen et al. (2023), this study employs the Random Forest (RF) and Extreme Gradient Boosting (XGBoost) models as the main machine learning model settings due to their best predictive performance on corporate bankruptcy prediction. Moreover, since the research sample firms have the characteristics of data imbalance (namely the number of non-bankrupt firms are much greater than that of bankrupt firms), this study also follows Chen et al. (2023) to employ the algorithms of EasyEnsemble, BalanceBaggingClassifier, and RandomUnderSampling to alleviate the limitation of data imbalance. Furthermore, to improve banks' decision-making effectiveness in loan practices, we especially focus on the improvements on the F1-score, AUC (Area Under the ROC Curve), MCC (Matthews Correlation Coefficient), the number of False Positive (FP), and the number of False Negative (FN).¹ It has to be noted that the reason why using the number of FN rather than Type I error is that the number of bankrupt firms is very small, resulting in large fluctuations on Type I error even if the number of FN changes slightly.

This study explores whether including earnings call transcripts *TCV* improves the bankruptcy prediction effectiveness of machine learning models using American bankrupt and non-bankrupt firm data from 2005 to 2020. The data of U.S. corporate bankruptcy are gathered from Capital IQ bankruptcy database. Since earnings call transcripts *TCV* variables can capture the forward-looking information about asset value distributions, this study expects additionally including earnings call transcripts *TCV* variables into the model setting of Barboza et al. (2017) perform better than the setting of Barboza et al. (2017). In addition, to

¹ In this study, Positive (Negative) is defined as firm bankruptcy (non-bankruptcy). Hence, False Positive represents an actual non-bankrupt firm and the prediction result is a bankrupt firm; False Negative represents an actual bankrupt firm and the prediction result is a non-bankrupt firm.

further explore whether earnings call transcripts *TCV* variables have different impacts on the term structure of bankruptcy risk, we employ the data of non-bankrupt firms and bankrupt firms from 2005 to 2016 as the training group to predict bankruptcy events in the future one, two, three, and four years in a rolling way of year by year. Moreover, to increase the robustness of the empirical results, this study (1) considers 30 kinds of random state settings for each prediction model and use the average of each model's performance evaluation indicators under 30 random state settings as the final prediction result; (2) considers annual report *TCV* variables (Chen et al. 2023) as additional model input variables and discusses whether earnings call transcripts *TCV* variables indeed provide different information from annual report *TCV* variables and still improve the model effectiveness of bankruptcy predictions; (3) introduces neural network models (e.g. DNN) as another model setting of bankruptcy prediction.

Empirical results of this study present that additionally including earnings call transcripts *TCV* variables significantly improve the bankruptcy prediction performance compared to the benchmark model settings of Barboza's et al. (2017) 11 financial variables (hereafter denoted as *Barboza_FIN*). Among the model performance indicators, significant improvements are observed in the F1-score, AUC, and MCC, particularly for the XGBoost model with EasyEnsemble algorithm. In this model setting, including earnings call transcripts *TCV* variables leads the F1-score (AUC; MCC) of the next one-year bankruptcy prediction to raise from 0.2806 to 0.3487 (0.7947 to 0.8213; 0.3022 to 0.3662), namely F1-score, AUC, and MCC significantly increase 6.81%, 2.66%, and 6.40%, respectively. The results imply that earnings call transcripts *TCV* variables can provide additional bankruptcy risk information beyond *Barboza_FIN* variables. Besides, compared to the model with *Barboza_FIN* variables, we also find that the number of False Positive (namely misjudges non-bankrupt firms as bankrupt ones) can be greatly reduced (i.e. from 254.1917 to 172.4250) while the number of False Negative (namely misjudges bankrupt firms as non-bankrupt ones) is slightly increased (i.e. from 7.3750 to 9.4667) after introducing earnings call transcripts *TCV* variables into bankruptcy prediction models. The economic implication is that the text-based communicative value of earnings call transcripts can effectively improve the situation where the model misjudges a non-bankrupt firm as a bankrupt firm. Therefore, earnings call transcripts *TCV* variables can improve the efficiency of capital utilization, effectively boost the operating performance of credit granting business, and further reduce the probability of credit contagion in the supply chain.

Meanwhile, the incremental improvement magnitude of earnings call transcripts *TCV* variables on bankruptcy prediction are pronounced for the future longer-term forecasting periods. Taking the XGBoost model with EasyEnsemble algorithm as example, including earnings call transcripts *TCV* variables leads the increase on the F1-score (AUC; MCC) of the next one-, two-, three-, and four-year bankruptcy predictions by 0.0681 (0.0266; 0.0640), 0.0715 (0.0326; 0.0679), 0.0749 (0.0377; 0.0728), and 0.0763 (0.0372; 0.0734), respectively. The results are consistent with the forward-looking characteristics of earnings call transcripts, which is complementary to that incomplete accounting information (e.g. annual report) is more pronounced for the shorter-term credit risk (Duffie and Lando 2001; Yu 2005; Chen and Tseng 2021; Chen et al. 2023). Moreover, the incremental improvement effect of earnings call transcripts *TCV* variables on bankruptcy prediction is still significant (1) when replacing RF and XGBoost machine learning models by neural network models

(e.g. DNN); (2) even when additionally introducing Chen et al.'s (2023) 11 annual report *TCV* (hereafter denoted as *Chen_TCV*) variables as another benchmark model input variables, especially for XGBoost model with EasyEnsemble and BalancedBagging algorithms. The robustness evidences support that earnings call transcripts *TCV* variables indeed provide the specific information for bankruptcy prediction distinct from annual report *TCV* variables.

Furthermore, to gain further insights into which earnings call transcripts *TCV* variables are crucial bankruptcy prediction indicators, this study conducts feature engineering on these proposed research variables. The empirical results suggest that the positive tone and uncertainty tone variables are the top two influential factors among earnings call transcripts *TCV* variables. Besides, these two features rank as the fourth to fifth (ninth to tenth) most important under the XGBoost (Random Forest) feature engineering when considering all variables, including the 11 financial variables proposed by Barboza et al. (2017), 11 annual report *TCV* variables proposed by Chen et al. (2023). For the variables of positive tone and uncertainty tone of earnings call transcripts, the former reflects the degree of optimistic expectations for the firm's future asset value distribution and the latter reflects the precision level of the communicative value of a firm's forward-looking information. Together, these variables not only describe the firm's future asset value distributions but also signal the firm's prospective financial condition, which, in turn, are closely linked to corporate bankruptcy risk in line with structural-form credit risk model frameworks (Merton 1974; Duffie and Lando 2001). In addition, to improve model interpretability, we employ Shapley Additive Explanations (SHAP; Lundberg and Lee 2017) to examine the contribution and directional impact of the selected earnings call transcripts *TCV* variables on the performance of the bankruptcy prediction model. The results show that both positive tone and uncertainty tone of earnings call transcripts exhibit relatively high and stable average absolute SHAP values, consistently ranking among the top six predictors for the next one- to four-year bankruptcy forecasts. Furthermore, we find that the positive tone variable is negatively related to corporate bankruptcy risk, while the uncertainty tone variable exerts the opposite effect, aligning with the theoretical expectations of structural-form credit risk models.

The incremental contributions include: (1) introducing earnings call transcripts *TCV* variables to bankruptcy prediction model and demonstrating the special features and economic implications of earnings call transcripts *TCV* variables compared to other textual variables of annual reports used in previous studies (Mai et al. 2019; Chen et al. 2023); (2) presenting the evidences that earnings call transcripts *TCV* variables still significantly improve the bankruptcy prediction performance (measured by F1-score, AUC, and MCC) in addition to considering 11 *Barboza_FIN* and 11 *Chen_TCV* variables; (3) presenting the evidences that earnings call transcripts *TCV* variables can significantly reduce the misjudgment of non-bankrupt firms as bankrupt ones (namely the number of False Positive), which effectively boost the operating performance of credit granting business; (4) the incremental improvement magnitude of earnings call transcripts *TCV* variables on bankruptcy prediction are pronounced for the future longer-term forecasting periods, consistent with the characteristics of the forward-looking information of earnings call transcripts and complementary to the findings of Chen et al. (2023) regarding annual report *TCV* variables; and (5) providing model interpretability through SHAP analyses, which highlight positive tone and uncertainty tone as the primary earnings call transcripts *TCV* variables driving corporate

bankruptcy, with the former negatively and the latter positively associated. Based on the above contributions, this study can conclude that earnings call transcripts *TCV* variables not only fill the academic gap in bankruptcy prediction literature but also are beneficial to the operational efficiency of credit granting business in loan practices.

The remainder of our study is organized as follows. Section 2 demonstrates the literature review. Section 3 presents the research methods, including data, research procedures, machine learning models, and evaluation indicators. Section 4 presents and analyzes empirical results. Finally, Sect. 5 provides concluding remarks.

2 Literature review

This study aims to explore whether incorporating earnings call transcripts *TCV* variables into bankruptcy prediction models using machine learning approaches can still improve the model effectiveness compared to the benchmark model with Barboza et al.'s (2017) 11 financial variables and even Chen et al.'s (2023) 11 annual report *TCV* variables. This section introduces the relevant literature from three aspects, including (1) the related researches on bankruptcy risk (including those using machine learning models), (2) the related researches on earnings call transcripts *TCV*, and (3) the researches on the relationship between financial report *TCV* and corporate credit/bankruptcy risk. The details are demonstrated as follows.

2.1 Researches on bankruptcy risk

Early research on bankruptcy prediction relied on statistical models using financial indicators, such as Beaver (1966), Altman's (1968) Z-Score, Ohlson's (1980) O-Score, and Zmijewski's (1984). As machine learning and AI technologies have advanced, many studies have incorporated machine learning models alongside financial accounting variables to predict corporate bankruptcy risk (e.g., Atiya 2001; Shin et al. 2005; Kumar and Ravi 2007; Tsai and Wu 2008).² Nanni and Lumini (2009) further find that machine learning models often outperform traditional statistical methods in bankruptcy prediction. While much research has focused on refining machine learning models using conventional financial variables (e.g., Altman 1968) or comparing model effectiveness, recent studies increasingly introduce new input variables. Examples include additional financial factors (Carton and Hofer 2006; Barboza et al. 2017), corporate governance (Liang et al. 2016), MD&A word frequencies (Mai et al. 2019), and annual report *TCV* features such as readability and tone (Chen et al. 2023). Despite these advancements, research on incorporating presentation and discussion sections of earnings call transcripts, along with the communicative value of managers' and analysts' expressions (such as readability and tone), remains limited. This study thus explores whether earnings call transcripts *TCV* variables can improve bankruptcy prediction effectiveness in current machine learning models.

² In addition to corporate bankruptcy prediction, consumer loan default prediction represents another important area of financial risk analysis. For instance, LaVoice and Vamossy (2024) examine racial disparities in debt collection, while Albanesi and Vamossy (2024) boost credit scoring performance to promote more equitable access to credit. Moreover, Albanesi and Vamossy (2019) discuss the application of deep learning techniques to improve consumer default prediction.

2.2 Researches on information contents of earnings call transcripts

2.2.1 Earnings call transcripts

An “earnings conference call” is a meeting convened by a company or organized through a brokerage firm, where top management team members reports the company’s operation results and future outlooks to institutional investors. During the earnings conference call, analysts and investors can also mention questions regarding the company’s financial performance and future operating perspectives. Therefore, earnings conference call provides a two-way communication platform between internal managers and external institutional investors. Beyond conveying more immediate, forward-looking, and relevant information about operations, investments, and financing activities to external investors, the discussion dialogues between analysts and top management team members during the earnings conference call can enhance external investors’ understanding of the company’s actual operation conditions and thus firm asset value distributions. The informational content of such discussion dialogues can also contributes to increasing the transparency of the company’s disclosures. That is, the textual information in the earnings call transcripts of operating outcomes presented by the top management team members, as well as the discussion dialogues (e.g. Q&A content) with external institutional investors, holds a certain level of informational content and communicative value. This study follows Seebeck and Kaya (2023) and Chen et al. (2023) to introduce the concepts of the text-based communication value and further employs the readability and tones variables of earnings call transcripts as the proxies of earnings call transcripts *TCV*. It has to be noted that the textual contents of earnings call transcripts include the presentations and discussion paragraphs, contributed by managers and analysts.

Owing to the timeliness and interactive nature of earnings call transcripts, they possess unique advantages over annual reports in conveying management’s sentiment and confidence about future business performance. Since these earnings call conferences are typically held shortly after the release of financial statements, managers must respond in real time to questions from investors and analysts, allowing managers’ verbal expressions, word choices, and tone to reveal attitudes toward future prospects (e.g. Price et al. 2012; Fu et al. 2021; Borochin et al. 2018).³ Hence, earnings call transcripts provide (1) immediate and responsive insights, (2) forward-looking signals through sentiment and tone, and (3) a perspective that contrasts with the inherent characteristics and constraints of annual reports, capturing subtle sentiments that are often difficult to detect in formal financial disclosures.

As the content of earnings call transcripts contains specific informational content (e.g., Matsumoto et al. 2011), especially in the presentation section and discussion section, it possesses more informational content compared to the accompanying press releases. Matsumoto et al. (2011) demonstrate that both the presentation and discussion sections have distinct informational content, with the discussion section exhibiting a higher level, attracting more analyst attention to the firm. Matsumoto et al. (2011) also further find that managers

³ Prior research shows that the tone and content of earnings calls contain predictive information about firms’ future performance, risks, and even crash risk, beyond what is available in financial statements (Price et al. 2012; Fu et al. 2021; Mamaysky et al. 2022). In addition, analysts’ questions during the Q&A sessions further elicit management’s perspectives on uncertainties and strategic outlooks (Borochin et al. 2018).

tend to provide increased disclosure in the presentation section of earnings call transcripts when a firm's operating performance is poor; however, in such circumstances, relatively more information is released in the discussion section. In addition, Arslan-Ayaydin et al. (2016) find that managers with equity-based incentives (such as stock options) are inclined to adopt a more exaggerated tone during earnings conference calls. Meanwhile, when managers' investment portfolio value is closely linked to the firm's stock price, the tone of earnings call transcripts tends to be more positive. Furthermore, Amoozegar et al. (2020) discover the tone of earnings call transcripts is significantly associated with the types of institutional investors. For example, the study finds that short-term institutional investors are positively associated with positive tones of earnings call transcripts while long-term institutional investors are positively associated with negative tones of earnings call transcripts. From the aforementioned findings, it is evident that earnings call transcript content, especially in the presentation and discussion sections, contains informational content, and the expressed tone is closely linked to managers' equity incentives and the types of institutional investors.

2.2.2 Economic consequences of information contents of earnings call transcripts

After discussing the significance of the information content and potential causes of the tones in earnings call transcripts, this study proceeds to explore the economic consequences of the information contents in earnings call transcripts. Numerous scholars have previously utilized textual features (such as tones) of earnings call transcripts to investigate their effects on stock price crash risk (Fu et al. 2021), abnormal stock returns and trading volume (Price et al. 2012), and firm value uncertainty (Borochin et al. 2018). Among these studies, Fu et al. (2021) find that firms with fewer positive tones in their earnings call transcripts are associated with a higher stock price crash risk in the following year. The impacts of tones of earnings call transcripts are more pronounced in better information environments or when there are fewer managerial incentive-based reasons. Price et al. (2012) identify that the tones of earnings call transcripts is an important predictor of abnormal stock returns and trading volume. In addition, the tones of earnings call transcripts plays a significant role in subsequent earnings surprises, and the discussion paragraphs in earnings call transcripts also affect post-announcement earnings changes. Borochin et al. (2018) discover that the analysts' tones in earnings calls has a more noticeable impact on firm value uncertainty (measured by implied volatility of stock options) compared to the managers' tones. Furthermore, the difference in tones between analysts and managers in their respective expressions is significantly associated with increased firm value uncertainty.

2.2.3 The association between information contents of earnings call transcripts and credit risk

Based upon the above studies, it is evident that the information content features in earnings call transcripts (e.g. readability and tones) have a certain association with firm value and its uncertainty. The textual information content in the presentation paragraphs by managers and the textual information content in the discussion paragraphs by managers and analysts

also hold their respective communicative values. Furthermore, since asset value, asset value volatility, and incomplete information are key theoretical components affecting firm credit risk (Merton 1974; Duffie and Lando 2001), it is reasonable to expect a certain association between the textual features of earnings call transcripts and corporate bankruptcy risk.

However, there is currently a scarcity of research that uses the textual features of earnings call transcripts as proxies for communicative value in exploring credit risk (e.g., Donovan et al. 2021). Moreover, literature combining the textual features of earnings call transcripts with the application of machine learning models for bankruptcy prediction is even rarer. This study aims to utilize various textual feature information (e.g. readability, tones) from the presentation paragraphs, discussion paragraphs, manager expressions, and analyst expressions in earnings call transcripts as proxies for the communicative value of earnings call transcripts. These will be employed as additional model input variables for the bankruptcy prediction model, and an analysis will be conducted to compare the effectiveness among various machine learning models.

It is essential to note that Donovan et al. (2021) utilize textual information from the MD&A section in annual reports and earnings call transcripts to predict CDS (credit default swap) spreads using machine learning models, creating a text-based credit risk indicator. However, CDS spreads differ from corporate bankruptcy events, and the aim of Donovan et al. (2021) is to establish a text-based credit risk indicator rather than using the information content of earnings call transcripts to improve the classification effectiveness of corporate bankruptcy prediction and provide further suggestions for credit loan strategies in practices.

In addition, unlike most studies employing machine learning models in bankruptcy prediction, this study not only provides a theoretical foundation based on structural-form credit risk models (Merton 1974; Duffie and Lando 2001) but also incorporates the concept of bankruptcy risk term structure. This inclusion offers financial institutions more flexibility in credit loan strategies. Therefore, in the context of utilizing machine learning models and integrating text-based communicative value features from earnings call transcripts in bankruptcy prediction, this research can be considered a pioneering study in this field.

2.3 Annual report textual characteristics and corporate credit/bankruptcy risk

Some previous studies have indicated a close association between annual report textual characteristics (e.g. readability, tones) and corporate credit risk, such as Ertugrul et al. (2017); Bonsall and Miller (2017); Chen and Tseng (2021); Chen et al. (2025b). Ertugrul et al. (2017) and Bonsall and Miller (2017) both present that firms with lower annual report readability have higher default risks and greater costs of debts. Chen and Tseng (2021); Chen et al. (2025b) demonstrate the associations from the perspectives of the readability of notes to consolidated financial statements and the readability of pension narrative disclosure, respectively.

Regarding the tone variables in annual reports, they serve as indicators of managers' opinions, sentiment, and attitudes toward current management outcomes and future prospects. According to Ertugrul et al. (2017), firm using more words with ambiguous tone and displaying weak modal words in their annual reports are associated with elevated credit

risks and more stringent loan contract terms. In light of the aforementioned discussions, we can conclude that the lower annual report readability, the increased tone ambiguity in annual reports, and a higher proportion of weak modal words in annual reports all contribute to higher default risk. In essence, the tone-related variables in annual reports significantly influence a firm's credit risk.

Recently, there has been a growing body of literature integrating annual report textual characteristics with machine learning models for corporate bankruptcy prediction, such as the word frequency in MD&A section (Mai et al. 2019) and annual report *TCV* variables (readability and tones) (Chen et al. 2023). However, unlike existing literature that predominantly focuses on annual report textual characteristics, this study introduces textual features of earnings call transcripts into the bankruptcy prediction model.

As mentioned in the previous section, earnings call transcripts offer real-time, forward-looking information relevant to operations, investments, and financing activities beyond the static attributes of management statements about financial performance found in annual reports. In addition to providing external investors with more immediate and pertinent information, conference call transcripts also supply dynamic information regarding analysts' and top management teams' discussions on the firm's future outlook and decision-making processes during earnings conference calls. Therefore, the information contained in earnings call transcripts possesses unique characteristics compared to annual reports.

In summary, this study aims to integrate earnings call transcripts *TCV* variables with machine learning models in the context of bankruptcy prediction research. First, the study validates whether incorporating these earnings call transcripts *TCV* variables into the Barboza et al.'s (2017) financial variable framework significantly enhances the effectiveness of corporate bankruptcy prediction. Second, the study then analyzes and compares the effectiveness of various machine learning models, incorporating considerations for the bankruptcy risk term structure. Lastly, the study expects to not only address academic gaps in bankruptcy prediction models but also offer strategic implications in credit loan practices by incorporating text-based communicative value characteristics derived from earnings call transcripts.

3 Research methodology

In this section, this study outlines the research methodology, covering aspects such as data and variables, data pre-processing, machine learning models, and evaluation methods and indicators. First, the study gathers data on U.S. bankrupt and non-bankrupt firms spanning from 2005 to 2020 shown in Compustat database and Capital IQ bankruptcy database. The study then implements data pre-processing procedures (e.g. excluding data with null values, removing outliers, and performing data standardization) and deals with data imbalance for bankruptcy prediction models. Second, this study partitions our sample into training and testing datasets based on time. The primary training data period spans from 2005 to 2016, while the testing data involves a rolling yearly approach from 2017 to 2020. Finally, the study assesses the prediction performance of the models using a confu-

sion matrix and related evaluation metrics such as F1-score, AUC, and MCC. The details are elaborated as follows.

3.1 Data and variables

3.1.1 Dependent variable (*Brupt*)

This study collects all U.S. bankrupt and non-bankrupt firm data from 2005 to 2020 as the preliminary research sample from Compustat database and Capital IQ bankruptcy database. The financial variable data (Barboza et al. 2017), earnings call transcripts data, and annual report *TCV* variables (Chen et al. 2023) are gathered from Compustat, Capital IQ, and SEC Analytics Suite databases, respectively. The study defines the *Brupt* variable as a dummy variable of bankrupt firms that labels bankrupt firms as 1 and non-bankrupt firms as 0. After data processing, the research dataset finally includes 1,045 bankrupt firms and 12,902 non-bankrupt firms.

3.1.2 Barboza et al.'s (2017) financial variables (*Barboza_FIN*)

Regarding the benchmark model setting, this study follows Barboza et al. (2017) to employ 11 financial variables as the basic model input variables, consisting of five component variables Altman's (1968) Z score and six financial changes variables introduced by Carton and Hofer (2006).⁴ The detailed definitions of these 11 financial variables are shown in Panel A of Table 1.

3.1.3 Chen et al.'s (2023) annual report *TCV* variables (*Chen_TCV*)

For the robustness purpose of validating that the information contents of earnings call transcripts *TCV* variables are different from those of annual report *TCV* variables, this study additionally introduces Chen et al.'s (2023) 11 annual report *TCV* variables as another benchmark model input variables. The details of these 11 annual report *TCV* variables are presented in Panel B of Table 1.

3.1.4 Earnings call transcripts *TCV* variables (*EC_TCV*)

As mentioned previously, the presentation and discussion paragraphs of earnings call transcripts have the distinct informational content, particularly evident in the discussion paragraphs (Matsumoto et al. 2011). In addition, Borochin et al. (2018) also find that the tone differences in the expressions of analysts and managers are significantly associated with the

⁴ Altman's (1968) Z score components include the following five variables: net working capital to total assets ratio (*NWC_TA*), retained earnings to total assets ratio (*RE_TA*), return on assets (*EBIT_TA*), equity market value to total debts ratio (*EMV_Debt*), and net sales to total assets ratio (*Sales_TA*). The six financial change variables proposed by Carton and Hofer (2006) include the following: earnings before interests and taxes to net sales ratio (*EBIT_Sales*), total assets growth rate (*TA_growth*), net sales growth rate (*Sales_growth*), number of employees growth rate (*EMP_growth*), return on equity change (*ROE_Chg*), and equity market-to-book value ratio change (*PB_Chg*).

Table 1 Variables definitions: Barboza et al. (2017) and Chen et al. (2023)

Variable	Definitions
Panel A. Barboza et al.'s (2017) financial variables	
NWC_TA	Net working capital divided by total assets
RE_TA	Retained earnings divided by total assets
EBIT_TA	Earnings before interests and taxes divided by assets
EMV_Debt	Equity market value divided by total debts
Sales_TA	Net sales divided by total assets
EBIT_Sales	Earnings before interests and taxes divided by net sales
TA_growth	The yearly growth rate of total assets
Sales_growth	The yearly growth rate of net sales
EMP_growth	The yearly growth rate of the number of employee
ROE_Chg	The annual change of return on equity
PB_Chg	The annual change of equity market-to-book value ratio
Panel B. Chen et al.'s (2023) annual report text-based communicative value (TCV) variables	
FSIZE	The file size of a firm's annual 10-K document (presented as megabytes)
FK_Ease	Flesch Reading Ease Index of a firm's annual report (10-K) $206.835 - 1.015(N_W/N_Sent) - 84.6(N_Syl/N_W)$ <i>[N_W, N_Sent, and N_Syl represent number of words, number of sentences, and number of syllables, respectively]</i>
CL	Coleman Readability Index of 10-K (Coleman and Liau 1975) $5.88(N_Char/N_W) - 29.6(N_Sent/N_W) - 15.8$ <i>[N_W, N_Sent, and N_Char represent number of words, number of sentences, and number of characters, respectively]</i>
A_WPP	Average number of words per paragraph of 10-K: N_W/N_Para <i>[N_W and N_Para represent number of words and number of paragraphs, respectively]</i>
FT_NEG	The percentage of negative words (Loughran-McDonald dictionary) in 10-K N_NegW_LM/N_W_LM <i>[N_NegW_LM represents the number of financial-negative words (Loughran-McDonald) in the annual report and N_W_LM is number of words that occur in the master dictionary]</i>
FT_MWeak_C	The number of count of modal weak words listed in the Loughran-McDonald dictionary in 10-K
FT_POS	The percentage of positive words (Loughran-McDonald dictionary) in 10-K N_PosW_LM/N_W_LM <i>[N_PosW_LM represents the number of financial-positive words (Loughran-McDonald) in the annual report and N_W_LM is number of words that occur in the master dictionary]</i>
Harv_NEG	The percentage of negative words (Harvard General Inquirer) in 10-K N_NegW_Harv/N_W_Harv <i>[N_NegW_Harv represents the number of negative words (Harvard General Inquirer) in the annual report and N_W_Harv is number of words that occur in Harvard General Inquirer]</i>
FT_LITI	The percentage of litigious words (Loughran-McDonald dictionary) in 10-K N_LitiW_LM/N_W_LM <i>[N_LitiW_LM represents the number of financial-litigious words (Loughran-McDonald) in the annual report and N_W_LM is number of words that occur in the master dictionary]</i>
FT_UNC	The percentage of uncertainty words (Loughran-McDonald dictionary) in 10-K N_UncW_LM/N_W_LM <i>[N_UncW_LM represents the number of financial-uncertainty words (Loughran-McDonald) in the annual report and N_W_LM is number of words that occur in the master dictionary]</i>

Table 1 (continued)

Variable	Definitions
FT_MStr	The percentage of financial-modal strong words (Loughran-McDonald dictionary) in 10-K N_MStrW_LM / N_W_LM <i>[N_MStrW_LM represents the number of financial-modal strong words (Loughran-McDonald) in the annual report and N_W_LM is number of words that occur in the master dictionary]</i>

increased uncertainty of firm value. Therefore, to capture the complete information contents of earnings call transcripts, this study adopts the full textual contents of earnings call transcripts as the primary source for estimating the newly proposed engineered NLP measures, namely earnings call transcripts *TCV* (*EC_TCV*) variables. The detailed definitions of these 27 *EC_TCV* variables are demonstrated in Appendix A and Table 2. It has to be noted that the employed texts of earnings call transcripts in this study include two sections of “Presentation” and “Discussion”.

Table 3 presents the sample distributions of bankrupt firms and non-bankrupt firms during the period from 2005 to 2020. There are totally 13,947 annual firm observations (non-bankrupt firms: 12,902 & bankrupt firms: 1,045) with available data of Barboza et al.’s (2017) 11 financial variables (namely *Barboza_FIN*), Chen et al.’s (2023) 11 annual report *TCV* variables (namely *Chen_TCV*), and 27 earnings call transcripts *TCV* variables (*EC_TCV*). This study employs the sample observations during period from 2005 to 2016 nnas the baseline training data set and those during the rolling period year by year from 2017 to 2020 as testing data set.

Table 4 presents the (pre-winsorized) descriptive statistics (e.g. mean, median, standard deviation (stdev), minimum, and maximum) of bankruptcy dummy variable (*Brupt*), *Barboza_FIN*, *Chen_TCV*, and *EC_TCV* variables, respectively. During the sample period, the average of *Brupt* variable is 0.0749, revealing that only 7.49% of the full sample firms are bankrupt and thus there exists a severe data imbalanced concern on our sample firms. In addition, the average of Full_FOG variable is 13.4025, indicating that the readability of earnings call transcripts is ideal (e.g. Li 2008) and corresponds to the educational level of “College freshman”. Also, the average of Full_Complex_words, Full_Lwords, and Full_Words are 1176.9790, 2221.9020, and 9336.6040, respectively, revealing that the proportion of complex words (long words) to total words in earnings call transcripts is on average 12.61% (23.80%). The above results suggest that the words appeared in earnings call transcripts are not all so approachable and easy to understand. The definitions of Barboza et al.’s (2017) financial variables, Chen et al.’s (2023) annual report *TCV* variables, the newly proposed *EC_TCV* variables can be referred to Tables 1 and 2, respectively.

3.2 Data pre-processing

3.2.1 Null value processing and data standardization

To mitigate the possible distortions arising from missing values and variations in the scale of input variables in the bankruptcy prediction models, this study only retains the annual firm observations with non-missing values of the input variables and introduces standard-

Table 2 Variables definitions: earnings transcripts text-based communicative value (*EC_TCV*) variables

Variable	Definitions
Panel A. Readability dimension	
Full_FK	The Flesch–Kincaid (FK, Flesch 1948) Grade Level of a firm's earnings call transcripts. The higher value of FK variable indicates the lower readability level. $0.39 (N_W/N_Sent) + 11.8 (N_Syl/N_W) - 15.59$ <i>[N_W, N_Sent, and N_Syl represent number of words, number of sentences, and number of syllables, respectively]</i>
Full_ARI	The automated readability index (ARI, Smith and Senter 1967) of a firm's earnings call transcripts. The higher value of ARI variable indicates the lower readability level. $4.71 (N_Char/N_W) + 0.5 (N_W/N_Sent) - 21.43$ <i>[N_W, N_Sent, and N_Char represent number of words, number of sentences, and number of characters, respectively]</i>
Full_CL	The Coleman Readability (CL, Coleman and Liau 1975) Index of a firm's earnings call transcripts. The higher value of CL variable indicates the lower readability level. $5.88(N_Char/N_W) - 29.6(N_Sent/N_W) - 15.8$ <i>[N_W, N_Sent, and N_Char represent number of words, number of sentences, and number of characters, respectively]</i>
Full_FK_Ease	The Flesch Reading Ease Index of a firm's earnings call transcripts $206.835 - 1.015(N_W/N_Sent) - 84.6(N_Syl/N_W)$ <i>[N_W, N_Sent, and N_Syl represent number of words, number of sentences, and number of syllables, respectively]</i>
Full_FOG	The Gunning FOG index (Gunning 1952) of a firm's earnings call transcripts. The higher value of FOG variable indicates the lower readability level. $0.4 [(N_W/N_Sent) + 100 (N_ComplexW/N_W)]$ <i>[N_W, N_Sent, and N_ComplexW represent number of words, number of sentences, and number of complex words, respectively]</i>
Full_LIX	The LIX index of a firm's earnings call transcripts. The higher value of LIX variable indicates the lower readability level. $(N_W/N_Periods) + 100 (N_LongW/N_W)$ <i>[N_W, N_Periods, and N_LongW represent number of words, the number of periods (defined by period, colon or capital first letter), and number of long words, respectively]</i>
Full_SMOG	The SMOG grade (McLaughlin 1969) of a firm's earnings call transcripts $1.0430 [N_PolySy + (30/N_Sent)]^{0.5} + 3.1291$ <i>[N_PolySy and N_Sent represent number of polysyllables and number of sentences, respectively]</i>
Full_RIX	The RIX index of a firm's earnings call transcripts. The higher value of LIX variable indicates the lower readability level. N_LongW/N_Sent <i>[N_Sent and N_LongW represent number of sentences and number of long words, respectively]</i>
Full_Dale	The Dale–Chall (Dale) readability score (Dale and Chall 1948) of a firm's earnings call transcripts. The higher value of Dale variable indicates the lower readability level. $0.1579 [100 (N_DiffW/N_W)] + 0.0496 (N_W/N_Sent)$ <i>[N_W, N_Sent, and N_DiffW represent number of words, number of sentences, and number of difficult words, respectively]</i>
Panel B. Text structure complexity dimension	
Full_Char_PW	The average number of characters per word of a firm's earnings call transcripts.
Full_Syll_PW	The average number of syllables per word of a firm's earnings call transcripts.
Full_WPS	The average number of words per sentence of a firm's earnings call transcripts.
Full_Chars	The total number of characters of a firm's earnings call transcripts.

Table 2 (continued)

Variable	Definitions
Full_Words	The total number of words (python-readability package based) of a firm's earnings call transcripts.
Full_Sents	The total number of sentences of a firm's earnings call transcripts.
Full_Lwords	The total number of long words of a firm's earnings call transcripts.
Full_Complex_words	The total number of complex words of a firm's earnings call transcripts.
Full_Twords	The total number of words (blank based) of a firm's earnings call transcripts.
Full_countedinMcD	The total number of words appeared in Loughran-McDonald master dictionary of a firm's earnings call transcripts.
Full_Sylls	The total number of syllables of a firm's earnings call transcripts.
Panel C. Tone or sentiment dimension	
Full_NEG	The percentage of negative words (Loughran-McDonald master dictionary) in a firm's earnings call transcripts. N_NegW_LM/N_W_LM [N_NegW_LM represents the number of financial-negative words (Loughran-McDonald) in the earnings call transcripts and N_W_LM is number of words that occur in the master dictionary]
Full_POS	The percentage of positive words (Loughran-McDonald master dictionary) in a firm's earnings call transcripts. N_PosW_LM/N_W_LM [N_PosW_LM represents the number of financial-positive words (Loughran-McDonald) in the earnings call transcripts and N_W_LM is number of words that occur in the master dictionary]
Full_UNC	The percentage of uncertainty words (Loughran-McDonald master dictionary) in a firm's earnings call transcripts. N_UncW_LM/N_W_LM [N_UncW_LM represents the number of financial-uncertainty words (Loughran-McDonald) in the earnings call transcripts and N_W_LM is number of words that occur in the master dictionary]
Full_LITI	The percentage of litigious words (Loughran-McDonald master dictionary) in a firm's earnings call transcripts. N_LitiW_LM/N_W_LM [N_LitiW_LM represents the number of financial-litigious words (Loughran-McDonald) in the earnings call transcripts and N_W_LM is number of words that occur in the master dictionary]
Full_MStr	The percentage of financial-modal strong words (Loughran-McDonald master dictionary) in a firm's earnings call transcripts. N_MStrW_LM/N_W_LM [N_MStrW_LM represents the number of financial-modal strong words (Loughran-McDonald) in the earnings call transcripts and N_W_LM is number of words that occur in the master dictionary]
Full_Mweak	The percentage of financial-modal weak words (Loughran-McDonald master dictionary) in a firm's earnings call transcripts. N_MWeakW_LM/N_W_LM [N_MWeakW_LM represents the number of financial-modal weak words (Loughran-McDonald) in the earnings call transcripts and N_W_LM is number of words that occur in the master dictionary]
Full_Constr	The percentage of constraint words (Loughran-McDonald master dictionary) in a firm's earnings call transcripts. $N_ConstrW_LM/N_W_LM$ [$N_ConstrW_LM$ represents the number of financial-constraint words (Loughran-McDonald) in the earnings call transcripts and N_W_LM is number of words that occur in the master dictionary]

Table 3 Sample distribution of bankrupt firms and non-bankrupt firms

Year	Non-Bankruptcy	Bankruptcy	Total	Percentage
2005	73	2	75	2.67%
2006	153	8	161	4.97%
2007	780	73	853	8.56%
2008	979	88	1,067	8.25%
2009	972	87	1,059	8.22%
2010	962	95	1,057	8.99%
2011	928	90	1,018	8.84%
2012	941	80	1,021	7.84%
2013	934	78	1,012	7.71%
2014	901	65	966	6.73%
2015	901	74	975	7.59%
2016	878	70	948	7.38%
2017	864	69	933	7.40%
2018	856	65	921	7.06%
2019	918	60	978	6.13%
2020	862	41	903	4.54%
Total	12,902	1,045	13,947	7.49%

ization method to transform the value of each input variable into a standard normal distribution with mean 0 and standard deviation 1, respectively. Therefore, the final set of sample observations guarantees the presence of non-null values for input variables and maintains consistency in variable scales.

3.2.2 Data splitting

This study adopts the methodology outlined by Chen et al. (2023) to implement a time-based approach for partitioning the research sample observations into training and testing data sets for the execution of machine learning models. The training data set is utilized for constructing effective machine learning models for bankruptcy prediction, while the testing data set is employed to validate the forecasting outcomes of the bankruptcy prediction models.

The baseline training data set encompasses sample observations from the period 2005 to 2016, while the testing data set comprises observations from a rolling yearly period spanning from 2017 to 2020. As mentioned in Chen et al. (2023), the rationale for employing time as the basis for sample data splitting lies in two key factors: (1) the time-varying nature of information pertaining to corporate financial structures; and (2) the facilitation of learning data properties in time series data, particularly when utilizing continuous time intervals for training machine models.

This study establishes the baseline training data set by utilizing the time span from 2005 to 2016, while the subsequent years, spanning from the next year to the next four years, form the sample periods for testing data. To determine these sample periods for training and testing data, this study employs a year-by-year rolling adjustment method. For instance, when conducting corporate bankruptcy prediction for the next one year (e.g., 2017, 2018, 2019, 2020), this study utilizes the year-by-year rolling baseline sample periods (i.e., 2005 to 2016, 2005 to 2017, 2005 to 2018, 2005 to 2019). This methodology is consistently applied for predicting bankruptcies in the next two, three, and four years.

Table 4 Summary statistics of major variables

Variables	Mean	Median	Stdev	Minimum	Maximum
Brupt	0.0749	0.0000	0.2633	0.0000	1.0000
Panel A. Barboza et al.'s (2017) Financial (<i>Barboza_FIN</i>) Variables					
NWC_TA	0.1757	0.1570	0.1848	-2.2462	0.9707
RE_TA	0.0586	0.2311	1.5527	-41.6227	6.6667
EBIT_TA	0.0832	0.0829	0.1180	-3.1439	1.2711
EMV_Debt	512.7175	4.1160	15101.6300	0.0000	1182308.0000
Sales_TA	0.9954	0.8115	0.7301	0.0021	13.1775
EBIT_Sales	0.0507	0.1075	2.8871	-328.7100	0.9479
TA_growth	0.0925	0.0452	0.3168	-0.8397	6.6285
Sales_growth	0.0726	0.0460	0.6917	-0.9684	58.0939
EMP_growth	0.0510	0.0178	0.6582	-0.9932	62.8190
ROE_Chg	0.0165	-0.0023	5.4122	-125.6969	388.6946
PB_Chg	-0.0800	0.0405	60.0690	-1853.0880	1917.4990
Panel B. Chen et al.'s (2023) Annual Report Text-based Communicative Value (<i>Chen_TCI</i>) Variables					
FSIZE	15,928,676	14,530,533	15,734,383	197,865	434,657,609
FK_Ease	0.4033	-6.5176	18.1257	-64.5713	39.9669
CL	20.7249	22.1571	3.6573	12.3565	31.8253
A_WPP	108.4388	108.6769	39.2402	29.9887	1378.2220
FT_NEG	0.0221	0.0153	0.0323	0.0067	0.5301
FT_MWeak_C	565.1722	510	344.7533	22.0000	8750.0000
FT_POS	0.0083	0.0082	0.0018	0.0027	0.0217
Harv_NEG	0.0496	0.0431	0.0317	0.0229	0.5421
FT_LITI	0.0166	0.0163	0.0049	0.0033	0.0684
FT_UNC	0.0102	0.0097	0.0025	0.0032	0.0247
FT_MStr	0.0020	0.0017	0.0011	0.0002	0.0109
Panel C. Earnings Call Transcripts Text-based Communicative Value (<i>EC_TCI</i>) Variables					
Full_FK	8.7783	8.7463	1.0239	5.0395	15.9845
Full_ARI	11.1660	11.1208	1.1686	6.9929	20.5313
Full_CL	5.0061	4.9697	0.6712	2.5716	9.1639
Full_FK_Ease	68.9982	69.0994	5.1056	42.3063	89.1433
Full_FOG	13.4025	13.3590	1.0859	9.4944	20.5959
Full_LIX	44.8377	44.7609	3.0112	32.1198	62.3039
Full_SMOG	11.8951	11.8705	0.7648	8.8662	15.7367
Full_RIX	5.0061	4.9697	0.6712	2.5716	9.1639
Full_Dale	4.7086	4.6938	0.4069	2.9675	6.4824
Full_Char_PW	4.7230	4.7194	0.1302	4.2340	5.5900
Full_Syll_PW	1.3809	1.3795	0.0508	1.1859	1.6413
Full_WPS	20.7017	20.5706	1.7036	15.0753	38.5042
Full_Chars	43898.0600	44824.0000	10641.4600	3704.0000	218247.0000
Full_Words	9336.6040	9521.2500	2359.4980	666.0000	49252.0000
Full_Sents	455.3373	460.0000	120.5777	27.0000	2773.0000
Full_Lwords	2221.9020	2256.2500	508.1979	226.0000	9620.0000
Full_Complex_words	1176.9790	1183.7500	286.5862	140.0000	5304.0000
Full_Twords	9124.5070	9306.5000	2292.3760	658.0000	47297.0000
Full_countedinMcD	7364.9150	7510.5000	1858.6470	550.0000	37538.0000
Full_Sylls	11688.8600	11915.0000	2835.9450	1025.0000	56801.0000
Full_NEG	80.0581	77.5000	25.0456	15.0000	216.7500
Full_POS	114.3761	110.5000	42.6200	2.0000	785.0000

Table 4 (continued)

Variables	Mean	Median	Stdev	Minimum	Maximum
Full_UNC	80.8823	79.7500	22.4760	9.0000	218.0000
Full_LITI	15.9970	12.7500	9.6476	6.0000	102.0000
Full_MStr	42.6057	41.0000	15.1692	2.0000	153.0000
Full_Mweak	38.0881	37.3333	11.5980	5.0000	129.0000
Full_Constr	9.7491	9.0000	4.2202	1.0000	49.0000

Note: Panels A, B, and C of Table 4 present the pre-winsorized descriptive statistics for the *Barboza_FIN*, *Chen_TCV*, and *EC_TCV* variables, respectively. The variable ‘Brupt’ is a binary indicator of bankruptcy status, with a value of 1 assigned to bankrupt firms and 0 to non-bankrupt firms. Tables 1 and 2 provide detailed definitions for the 11 *Barboza_FIN* variables, 11 *Chen_TCV* variables, and 27 *EC_TCV* variables

3.2.3 Imbalanced data processing

Due to the prevalent imbalance in the distribution of data between bankrupt and non-bankrupt firms, addressing imbalanced data has become a crucial aspect in the realm of bankruptcy prediction research. For instance, the dataset used in this study comprises 12,902 non-bankrupt firms and 1,045 bankrupt firms. This indicates that a mere 7.49% of the entire sample consists of bankrupt firms, highlighting the pronounced issue of data imbalance in this investigation. To tackle this challenge, we adopt the approach proposed by Chen et al. (2023), employing EasyEnsemble and BalanceBaggingClassifier algorithms. The details of these two methods for addressing data imbalance are demonstrated in Appendix B.

3.3 Machine learning models

Based on the empirical findings of Chen et al. (2023), it has been determined that Random Forest (RF) and Extreme Gradient Boosting (XGBoost) algorithms exhibit superior predictive power compared to various other machine learning models. Consequently, in alignment with Chen et al. (2023), this study adopts Random Forest and XGBoost algorithms as our primary machine learning models. The specifics of these two algorithms are demonstrated below.

3.3.1 Random forest

The Random Forest algorithm is a type of ensemble learning model that consists of multiple decision trees, namely a Bagging model. The ultimate prediction outcome is determined through a voting mechanism (Breiman 2001). The decision-making process of the Random Forest algorithm involves several steps: (1) computing the information amount (Entropy) at each node within each level of the tree; (2) calculating the information gain (IG) by deducting the weighted average information of the nodes from the classified information; (3) choosing the feature with the highest information gain to serve as the basis for categorization. The characteristics of the Random Forest model are illustrated as follows: (1) greater efficiency compared to decision trees, especially in managing missing values and outliers; (2) reduced susceptibility to overfitting issues; (3) less restricted by the need for fine-tuning hyperparameters to make predictions.

3.3.2 XGBoost

The Extreme Gradient Boosting (XGBoost) algorithm, introduced by Chen and Guestrin (2016), is an extension and enhancement of the Gradient Boosted Decision Tree (GBDT) model. Rooted in the Boosting algorithm, XGBoost works by assembling numerous weak classifiers into a robust classifier. Throughout the training process, the algorithm amplifies the data error weight of the existing classifier and continues to train new classifiers. This iterative approach allows the new classifier to learn the characteristics of misclassified data, leading to performance improvement in subsequent training iterations. In addition, since XGBoost is based on the GBDT model, it addresses supervised learning problems through two key mechanisms: (1) amalgamating numerous tree models into a potent classifier, and (2) employing gradient descent to minimize residuals during the decision-making process. Finally, it has to be noted that the XGBoost algorithm also incorporates a regularization term as a penalty to address the potential overfitting issues.

3.4 Confusion matrix

This study follows Chen et al. (2023) to introduce the confusion matrix to assess the model effectiveness of the classification and prediction results. Within the confusion matrix, prediction outcomes are categorized into four groups: True Positive (TP), False Positive (FP), True Negative (TN), and False Negative (FN). This study designates Positive (Negative) as firm bankruptcy (non-bankruptcy)⁵. Evaluation metrics associated with the confusion matrix include Accuracy, F1-Score, AUCs (Area Under the ROC Curve), MCC (Matthews Correlation Coefficient), the number of False Negative (i.e. Type I error), and the number of False Positive (i.e. Type II error). The F1-Score, presented in Eq. (1), serves as the harmonic mean of the precision rate and recall rate. The precision rate signifies the proportion of correctly predicted positive (negative) samples among those identified as positive (negative) by the model, namely $\frac{TP}{TP+FP}$ ($\frac{TN}{TN+FN}$). The recall rate quantifies the proportion of actual positive (negative) samples that the model successfully predicts, namely $\frac{TP}{TP+FN}$ ($\frac{TN}{TN+FP}$). Moreover, this study defines the number of False Negative (False Positive) as the number of bankrupt (non-bankrupt) firm samples are misjudged as non-bankrupt (bankrupt) firms and thus Type I error (Type II error) is described by the ratio of the number of False Negative (False Positive) to the actual Positive (actual Negative) firm sample, namely $\frac{FN}{TP+FN}$ ($\frac{FP}{TN+FP}$).

$$F1 - Score = 2 * \frac{Precision * Recall}{Precision + Recall} \quad (1)$$

Furthermore, the Matthews Correlation Coefficient (MCC, Matthews 1975), defined as Eq. (2), is a metric used to assess the performance of binary classification models, particularly in cases of imbalanced datasets. The MCC score ranges from -1 to 1, with 1 repre-

⁵ In this study, since Positive means bankruptcy and Negative means non-bankruptcy, True Positive means an actual bankrupt firm and the prediction result is also bankrupt firm; True Negative represents an actual non-bankrupt firm and the prediction result is also non-bankrupt firm; False Positive represents an actual non-bankrupt firm and the prediction result is a bankrupt firm; False Negative represents an actual bankrupt firm and the prediction result is a non-bankrupt firm.

sending perfect predictions, 0 indicating random predictions, and -1 signifying completely inverse predictions. A higher MCC, closer to 1, reflects stronger predictive performance.

$$\text{MCC} = \frac{\text{TP} \times \text{TN} - \text{FP} \times \text{FN}}{\sqrt{(\text{TP} + \text{EP})(\text{TP} + \text{FN})(\text{TN} + \text{FP})(\text{TN} + \text{FN})}} \quad (2)$$

Compared to the financial variables proposed by Barboza et al. (2017) and even the annual report *TCV* variables introduced by Chen et al. (2023) as model input features, this study anticipates that the inclusion of earnings call transcripts *TCV* variables will improve the accuracy, F1-Score, AUC, and MCC of the machine learning models. Simultaneously, this inclusion is expected to decrease the occurrences of False Negative (i.e., Type I error) and False Positive (i.e., Type II error).

4 Empirical analyses

This section illustrates the empirical results of bankruptcy predictions through the utilization of two distinct machine learning models (Random Forest and XGBoost) combined with various data imbalance techniques (BalanceBaggingClassifier, EasyEnsemble, and RandomUnderSampling). This study explores whether including earnings call transcripts *TCV* variables enhances the effectiveness of bankruptcy prediction under the benchmark model settings based on Barboza et al. (2017) and Chen et al. (2023). For the robustness purpose, this study conducts analyses on the effectiveness of bankruptcy prediction across 30 randomly selected states for each machine learning model. The average values of each evaluation metric are calculated across these 30 states, finally yielding the results on the predictive effectiveness.

4.1 Comparative analyses of the effectiveness of bankruptcy prediction models before and after adding the earnings call transcripts *TCV* features

Table 5 displays the outcomes of bankruptcy prediction effectiveness over the subsequent year using Random Forest (RF) and XGBoost models, both before and after the inclusion of earnings call transcripts *TCV* (namely *EC_TCV*) variables. The benchmark training period spans from 2005 to 2016, and the training period is dynamically adjusted on a rolling basis each year to predict corporate bankruptcy events in the following year. The values presented in Table 5 represent the averages derived from the rolling forecast results for each year (specifically 2017, 2018, 2019, and 2020) across 30 distinct sets of random states.

Panel A in Table 5 shows the prediction results using RF and XGBoost models with *Barboza_FIN* variables, and Panel B presents the results incorporating both *Barboza_FIN* and earnings call transcripts *TCV* (*EC_TCV*) variables. The empirical findings in Table 5 indicate that, regardless of BalancedBagging and EasyEnsemble algorithms, the inclusion of earnings call transcripts *TCV* variables significantly improves the effectiveness of bankruptcy prediction models. For instance, in the case of the XGBoost model with EasyEnsemble (BalancedBagging) algorithms, additionally incorporating earnings call transcripts

Table 5 Comparative analyses of the effectiveness of bankruptcy prediction models (Forecasting period is the next one year)

Panel A. Bankruptcy Prediction using <i>Barboza_FIN</i> (2017) variables										
	Accuracy	AUC	F1-score	TP	TN	FN	FP	MCC	Type I error	Type II error
A.1. BalancedBagging										
RF	0.7915	0.7991	0.3252	47.1667	691.9583	11.5833	183.0417	0.3344	0.1926	0.2092
XG-Boost	0.7958	0.8016	0.3297	47.2833	695.8750	11.4667	179.1250	0.3400	0.1920	0.2047
A.2. EasyEnsemble										
RF	0.7051	0.7885	0.2716	51.6000	606.7416	7.1500	268.2583	0.2925	0.1166	0.3065
XG-Boost	0.7199	0.7947	0.2806	51.3750	620.8083	7.3750	254.1917	0.3022	0.1202	0.2904
A.3. RandomUnderSampling										
RF	0.7118	0.7783	0.2696	49.9000	614.7416	8.8500	260.2583	0.2845	0.1460	0.2974
XG-Boost	0.7000	0.7589	0.2553	48.3333	605.2500	10.4167	269.7500	0.2636	0.1740	0.3083
Panel B. Bankruptcy Prediction using <i>Barboza_FIN</i> (2017) and <i>EC_TCV</i> variables										
B.1. BalancedBagging										
RF	0.8371	0.8023	0.3676	44.2667	737.6667	14.4833	137.3333	0.3660	0.2384	0.1570
XG-Boost	0.8587	0.8161	0.4023	44.7583	757.3000	13.9917	117.7000	0.4018	0.2332	0.1346
B.2. EasyEnsemble										
RF	0.7715	0.8020	0.3137	48.7750	671.8167	9.9750	203.1833	0.3271	0.1638	0.2322
XG-Boost	0.8052	0.8213	0.3487	49.2833	702.5750	9.4667	172.4250	0.3662	0.1604	0.1970
B.3. RandomUnderSampling										
RF	0.7716	0.7897	0.3072	47.3333	673.3834	11.4167	201.6167	0.3159	0.1902	0.2304
XG-Boost	0.7843	0.7921	0.3163	46.9917	685.4250	11.7583	189.5750	0.3256	0.1990	0.2167

Note: AUC, F1-score, and MCC are the primary performance metrics used to evaluate the model. Definitions for TP, TN, FP, and FN can be found in the confusion matrix. Given the relatively small number of bankrupt firms, this study applies various algorithms to address data imbalance, including BalancedBagging, EasyEnsemble, and RandomUnderSampling

TCV variables results in an increase in accuracy from 0.7199 to 0.8052 (0.7958 to 0.8587). Moreover, the inclusion leads to an improvement in AUCs from 0.7947 to 0.8213 (0.8016 to 0.8161), an increase in F1-score from 0.2806 to 0.3487 (0.3297 to 0.4023), an increase in MCC from 0.3022 to 0.3662 (0.3400 to 0.4018), a substantial decrease in the number of Type II errors (FP) from 254.1917 to 172.4250 (179.1250 to 117.7000), and a slight increase in the number of Type I errors (FN) from 7.3750 to 9.4667 (11.4667 to 13.9917). The results also suggest that earnings call transcripts *TCV* variables can (1) lead accuracy, F1-score, AUC, and MCC to have a substantial increase; (2) effectively reduce the number of Type II error and keep the change on the number of Type I error at a relatively small level; (3) can provide additional bankruptcy risk information beyond *Barboza_FIN* variables.

Across the various model settings outlined in Table 5, the prediction AUCs has an average increase ranging from 0.32% to 3.33%; the F1-score has an average increase from

3.76% to 7.26%; the MCC has an average increase from 3.13% to 6.40%; the number of Type II error (FP) has an average decrease from 45.7084 to 81.7667; the number of Type I error (FN) has an average increase from 1.3417 to 2.9000. The above model evaluation results reveal that the earnings call transcripts *TCV* variables still indeed improve the forecasting effectiveness of bankruptcy prediction models even though these models have considered Barboza et al.'s (2017) 11 financial variables. Specifically, the XGboost model with BalancedBagging algorithm exhibits the best performance based on the criteria of F1 score (0.4023) and MCC (0.4018), whereas the XGBoost model with EasyEnsemble algorithm excels based on the AUCs criterion (0.8213).

In addition, this study also finds that the inclusion of earnings call transcripts *TCV* variables also results in a notable reduction in the number of Type II errors (FP) across various model settings, while maintaining a relatively modest impact on the number of Type I errors (FN). The substantial decrease in Type II errors is a crucial factor contributing to the improvement of model prediction effectiveness. Hence, earnings call transcripts *TCV* variables can greatly diminish the likelihood of misjudging a non-bankrupt firm as a bankrupt firm. The practical implications of these findings suggest that earnings call transcripts *TCV* variables have the potential to increase the probability of banks granting credit loans to customers with sound solvency profiles. This, in turn, can enhance the efficiency of capital utilization, effectively boost the operating performance of credit granting business, and mitigate the likelihood of credit contagion within the supply chain.

Panel A (Panel B) in Tables 6 and 7, and 8 demonstrate the results of bankruptcy prediction using RF and XGBoost models in the next two, three, and four years before (after) incorporating the earnings call transcripts *TCV* variables in addition to with *Barboza_FIN* variables, respectively. For the next two-year (three-year; four-year) bankruptcy prediction, the training period from 2005 to 2016, 2005 to 2017, and 2005 to 2018 (2005 to 2016 and 2005 to 2017; 2005 to 2016) are respectively used to predict corporate bankruptcy in 2017–2018, 2018–2019, and 2019–2020 (2017–2019 and 2018–2020; 2017–2020). Similarly, the values presented in these tables represent the averages derived from the rolling prediction results for each year across 30 distinct sets of random states.

The results of Tables 6 and 7, and 8 present that the prediction effectiveness of Random Forest and XGboost models has the significant improvement using the data imbalance processing methods of BalancedBagging and EasyEnsemble algorithms, and the number of Type II error has a great reduction while the number of Type I error has a slight increase after including the earnings call transcripts *TCV* variables. In addition, among these model settings, we also find that XGBoost models under BalancedBagging (EasyEnsemble) algorithm perform the best for the next two, three, and four forecasting years based on the criteria of F1 score (AUCs; MCC), namely 0.4038, 0.3938, and 0.3829 (0.8199, 0.8171, and 0.8089; 0.3999, 0.3871, 0.3743). Besides, the incremental improvements on bankruptcy effectiveness in the next two, three, and four years for the model settings including earnings call transcripts *TCV* variables under the XGBoost models with various data imbalanced algorithms are all greater than those under RF models. These results are consistent with the bankruptcy prediction result in the next one year (namely Table 5).

Based on the above results, this study therefore verifies that earnings call transcripts *TCV* variables (1) indeed improve the effectiveness of bankruptcy prediction models (especially the significant reduction of the Type II errors) in the next one, two, three, and four years,

Table 6 Comparative analyses of the effectiveness of bankruptcy prediction models (Forecasting period is the next two years)

Panel A. Bankruptcy Prediction using <i>Barboza_FIN</i> (2017) variables										
	Accuracy	AUC	F1-score	TP	TN	FN	FP	MCC	Type I error	Type II error
A.1. BalancedBagging										
RF	0.7867	0.7977	0.3258	96.9111	1380.3780	23.0889	377.6222	0.3336	0.1899	0.2147
XG-Boost	0.7916	0.7994	0.3303	96.7222	1389.9110	23.2778	368.0889	0.3381	0.1920	0.2093
A.2. EasyEnsemble										
RF	0.6950	0.7804	0.2681	105.0333	1200.0000	14.9667	558.0000	0.2855	0.1219	0.3173
XG-Boost	0.7115	0.7874	0.2781	104.5444	1231.5560	15.4556	526.4445	0.2960	0.1259	0.2994
A.3. RandomUnderSampling										
RF	0.7007	0.7709	0.2660	101.8333	1214.0220	18.1667	543.9778	0.2776	0.1488	0.3093
XG-Boost	0.6925	0.7543	0.2544	98.6778	1201.7780	21.3222	556.2222	0.2596	0.1750	0.3164
Panel B. Bankruptcy Prediction using <i>Barboza_FIN</i> (2017) and <i>EC_TCV</i> variables										
B.1. BalancedBagging										
RF	0.8363	0.8033	0.3726	91.1333	1479.4220	28.8667	278.5778	0.3696	0.2352	0.1583
XG-Boost	0.8569	0.8141	0.4038	91.1333	1518.1330	28.8667	239.8667	0.3999	0.2355	0.1363
B.2. EasyEnsemble										
RF	0.7676	0.7978	0.3129	99.3556	1342.1440	20.6444	415.8556	0.3242	0.1680	0.2364
XG-Boost	0.8010	0.8199	0.3496	100.7222	1403.5000	19.2778	354.5000	0.3639	0.1586	0.2015
B.3. RandomUnderSampling										
RF	0.7685	0.7878	0.3082	96.7333	1346.3670	23.2667	411.6333	0.3148	0.1905	0.2340
XG-Boost	0.7812	0.7902	0.3176	95.7778	1371.2780	24.2222	386.7222	0.3236	0.1997	0.2199

Note: AUC, F1-score, and MCC are the primary performance metrics used to evaluate the model. Definitions for TP, TN, FP, and FN can be found in the confusion matrix. Given the relatively small number of bankrupt firms, this study applies various algorithms to address data imbalance, including BalancedBagging, EasyEnsemble, and RandomUnderSampling

namely bankruptcy risk term structure; (2) increase the opportunity for banks to grant credit loans to customers with sound solvency ability, thereby improving the efficiency of capital utilization in bank loan practices and reducing the probability of credit contagion in the supply chain.

To provide more robust evidences for the importance of including the earnings call transcripts *TCV* variables on the effectiveness of bankruptcy prediction models, this study performs the mean differences tests of AUCs, F1 scores, and MCCs using the two samples of the machine learning models (namely Random Forest and XGBoost) with 30 random states before and after including the earnings call transcripts *TCV* variables. The results of mean difference tests are shown in Table 9. For each column of Table 9, (1) “Mean_Before” represents the average performance of the model setting with Barboza et al.’s (2017) financial variables; (2) “Mean_After” stands for the average performance of the model setting including *EC_TCV* variables in addition to Barboza et al.’s (2017) financial variables; (3) “Dif-

Table 7 Comparative analyses of the effectiveness of bankruptcy prediction models: (Forecasting period is the next three years)

Panel A. Bankruptcy Prediction using <i>Barboza_FIN</i> (2017) variables										
	Accuracy	AUC	F1-score	TP	TN	FN	FP	MCC	Type I error	Type II error
A.1. BalancedBagging										
RF	0.7830	0.7940	0.3216	144.7667	2061.1830	35.2333	575.8167	0.3284	0.1937	0.2184
XG-	0.7889	0.7968	0.3274	144.7500	2077.7000	35.2500	559.3000	0.3344	0.1942	0.2121
Boost										
A.2. EasyEnsemble										
RF	0.6899	0.7759	0.2645	157.1167	1786.4330	22.8833	850.5667	0.2808	0.1257	0.3226
XG-	0.7048	0.7794	0.2719	155.3333	1830.1500	24.6667	806.8500	0.2872	0.1353	0.3060
Boost										
A.3. RandomUnderSampling										
RF	0.6945	0.7650	0.2612	151.9167	1804.5170	28.0833	832.4833	0.2710	0.1543	0.3157
XG-	0.6870	0.7478	0.2498	146.8333	1788.4670	33.1667	848.5333	0.2527	0.1827	0.3218
Boost										
Panel B. Bankruptcy Prediction using <i>Barboza_FIN</i> (2017) and <i>EC_TCV</i> variables										
B.1. BalancedBagging										
RF	0.8340	0.7934	0.3647	133.9167	2215.6500	46.0833	421.3500	0.3592	0.2533	0.1598
XG-	0.8538	0.8033	0.3938	133.7333	2271.5000	46.2667	365.5000	0.3871	0.2547	0.1386
Boost										
B.2. EasyEnsemble										
RF	0.7657	0.7919	0.3092	147.5167	2009.6170	32.4833	627.3834	0.3185	0.1784	0.2379
XG-	0.7987	0.8171	0.3468	150.5500	2099.5170	29.4500	537.4833	0.3601	0.1620	0.2038
Boost										
B.3. RandomUnderSampling										
RF	0.7647	0.7805	0.3024	143.3500	2010.9330	36.6500	626.0667	0.3068	0.2015	0.2374
XG-	0.7767	0.7831	0.3111	141.9500	2046.0170	38.0500	590.9833	0.3147	0.2098	0.2241
Boost										

Note: AUC, F1-score, and MCC are the primary performance metrics used to evaluate the model. Definitions for TP, TN, FP, and FN can be found in the confusion matrix. Given the relatively small number of bankrupt firms, this study applies various algorithms to address data imbalance, including BalancedBagging, EasyEnsemble, and RandomUnderSampling

ference” stands for Mean_After minus Mean_Before; and (4) the T-test statistic represents the difference tests of the AUCs, F1-scores, and MCCs of the rolling forecast results of each year in the above two groups of input variables under 30 random states.

The results of the mean difference tests for AUCs, F1 scores, and MCCs in Table 9 show that the earnings call transcripts *TCV* variables (1) can significantly increase the F1 scores and MCCs of all six various model settings (RF and XGBoost models with three algorithms of BalancedBagging, EasyEnsemble, and RandomUnderSampling); (2) can significantly increase the AUCs of all six various model settings besides the RF model with the BalancedBagging algorithm; (3) have the greatest increase on the AUCs of the XGBoost model with the EasyEnsemble algorithm (ranging from 0.0266 to 0.0377), especially for future long-term periods (e.g. future 2, 3 and 4 years); (4) have the greatest increase on the F1 scores of the XGBoost models with the EasyEnsemble algorithm (ranging from 0.0681 to 0.0763), especially for future long-term periods (e.g. future 2, 3 and 4 years); (5) have the greatest increase on the MCCs of the XGBoost models with the EasyEnsemble algorithm

Table 8 Comparative analyses of the effectiveness of bankruptcy prediction models: (Forecasting period is the next four years)

Panel A. Bankruptcy Prediction using <i>Barboza_FIN</i> (2017) variables										
	Accuracy	AUC	F1-score	TP	TN	FN	FP	MCC	Type I error	Type II error
A.1. BalancedBagging										
RF	0.7805	0.7854	0.3120	185.8667	2729.4000	49.1333	770.6000	0.3175	0.2091	0.2202
XG-Boost	0.7865	0.7885	0.3179	185.8667	2751.5330	49.1333	748.4667	0.3235	0.2091	0.2138
A.2. EasyEnsemble										
RF	0.6892	0.7730	0.2602	204.1667	2369.9330	30.8333	1130.0670	0.2766	0.1312	0.3229
XG-Boost	0.7001	0.7716	0.2637	200.5667	2414.3670	34.4333	1085.6330	0.2776	0.1465	0.3102
A.3. RandomUnderSampling										
RF	0.6922	0.7594	0.2551	196.5000	2388.8670	38.5000	1111.1330	0.2640	0.1638	0.3175
XG-Boost	0.6810	0.7387	0.2411	189.1000	2354.5000	45.9000	1145.5000	0.2419	0.1953	0.3273
Panel B. Bankruptcy Prediction using <i>Barboza_FIN</i> (2017) and <i>EC_TCV</i> variables										
B.1. BalancedBagging										
RF	0.8329	0.7820	0.3528	170.1000	2940.9000	64.9000	559.1000	0.3455	0.2762	0.1597
XG-Boost	0.8533	0.7927	0.3829	170.0000	3017.0670	65.0000	482.9333	0.3743	0.2766	0.1380
B.2. EasyEnsemble										
RF	0.7648	0.7830	0.3007	188.9000	2667.4670	46.1000	832.5333	0.3083	0.1962	0.2379
XG-Boost	0.7999	0.8089	0.3400	192.5000	2795.0000	42.5000	705.0000	0.3511	0.1809	0.2014
B.3. RandomUnderSampling										
RF	0.7634	0.7720	0.2940	183.7333	2667.5330	51.2667	832.4667	0.2968	0.2182	0.2378
XG-Boost	0.7757	0.7741	0.3025	181.4667	2715.8330	53.5333	784.1667	0.3040	0.2278	0.2240

Note: AUC, F1-score, and MCC are the primary performance metrics used to evaluate the model. Definitions for TP, TN, FP, and FN can be found in the confusion matrix. Given the relatively small number of bankrupt firms, this study applies various algorithms to address data imbalance, including BalancedBagging, EasyEnsemble, and RandomUnderSampling

(ranging from 0.0640 to 0.0734), especially for future long-term periods (e.g. future 2, 3 and 4 years).⁶ Based on the above results, this study therefore concludes that the earnings call transcripts *TCV* variables provide the significantly incremental improvement, especially for the model setting of XGBoost model. In addition, we also find that including the earnings call transcripts *TCV* variables provide a certainly higher level of incremental predictive effectiveness (measured by AUCs, F1 score, and MCC) in the longer-term forecasting periods for the model settings with EasyEnsemble and RandomUnderSampling algorithms. The results reveal that the incremental impacts of the earnings call transcripts *TCV* variables

⁶ It has to be noted that the earnings call transcripts *TCV* variables have the increase on the AUCs of the XGBoost models with the BalancedBagging (ranging from 0.0042 to 0.0147) and RandomUnderSampling (ranging from 0.0333 to 0.0359) algorithms. Meanwhile, the earnings call transcripts *TCV* variables also have the increase on the F1 scores of the XGBoost models with the BalancedBagging (ranging from 0.0650 to 0.0735) and RandomUnderSampling (ranging from 0.0610 to 0.0632) algorithms. Similarly, there has a significant increase on the MCCs of the XGBoost models with the BalancedBagging and RandomUnderSampling algorithms.

Table 9 The difference tests analyses in the effectiveness of bankruptcy prediction models: before and after introducing earnings call transcripts TCV variables**Panel A. Forecasting Period is the Next One Year**

Model	Measure	Mean Before	Mean_After	Difference	T statistic	p-value
A.1. BalancedBagging						
RF	AUC	0.7991	0.8023	0.0032	1.50	0.1354
	F1-score	0.3252	0.3676	0.0425	21.90	0.0000
	MCC	0.3344	0.3660	0.0315	13.79	0.0000
	FP	183.0417	137.3333	-45.7084	-49.35	0.0000
	FN	11.5833	14.4833	2.9000	9.89	0.0000
XGBoost	AUC	0.8016	0.8161	0.0145	8.20	0.0000
	F1-score	0.3297	0.4023	0.0726	39.86	0.0000
	MCC	0.3400	0.4018	0.0618	28.23	0.0000
	FP	179.1250	117.7000	-61.4250	-114.64	0.0000
	FN	11.4667	13.9917	2.5250	11.02	0.0000
A.2. EasyEnsemble						
RF	AUC	0.7885	0.8020	0.0135	9.62	0.0000
	F1-score	0.2716	0.3137	0.0421	30.65	0.0000
	MCC	0.2925	0.3271	0.0346	21.46	0.0000
	FP	268.2583	203.1833	-65.0750	-50.47	0.0000
	FN	7.1500	9.9750	2.8250	17.02	0.0000
XGBoost	AUC	0.7947	0.8213	0.0266	11.89	0.0000
	F1-score	0.2806	0.3487	0.0681	40.77	0.0000
	MCC	0.3022	0.3662	0.0640	25.61	0.0000
	FP	254.1917	172.4250	-81.7667	-164.38	0.0000
	FN	7.3750	9.4667	2.0917	9.47	0.0000
A.3. RandomUnderSampling						
RF	AUC	0.7783	0.7897	0.0114	4.99	0.0000
	F1-score	0.2696	0.3072	0.0376	19.10	0.0000
	MCC	0.2845	0.3159	0.0313	12.19	0.0000
	FP	260.2583	201.6167	-58.6416	-35.17	0.0000
	FN	8.8500	11.4167	2.5667	10.89	0.0000
XGBoost	AUC	0.7589	0.7921	0.0333	13.84	0.0000
	F1-score	0.2553	0.3163	0.0610	30.17	0.0000
	MCC	0.2636	0.3256	0.0620	22.06	0.0000
	FP	269.7500	189.5750	-80.1750	-53.73	0.0000
	FN	10.4167	11.7583	1.3417	5.46	0.0000

Panel B. Forecasting Period is the Next Two Years

Model	Measure	Mean Before	Mean_After	Difference	T statistic	p-value
B.1. BalancedBagging						
RF	AUC	0.7977	0.8033	0.0055	4.77	0.0000
	F1-score	0.3258	0.3726	0.0468	64.65	0.0000
	MCC	0.3336	0.3696	0.0360	40.88	0.0000
	FP	377.6222	278.5778	-99.0444	-73.12	0.0000
	FN	23.0889	28.8667	5.7778	14.46	0.0000
XGBoost	AUC	0.7994	0.8141	0.0147	9.77	0.0000
	F1-score	0.3303	0.4038	0.0735	72.76	0.0000
	MCC	0.3381	0.3999	0.0618	44.27	0.0000
	FP	368.0889	239.8667	-128.2222	-186.22	0.0000
	FN	23.2778	28.8667	5.5889	13.28	0.0000

Table 9 (continued)**Panel A. Forecasting Period is the Next One Year****B.2. EasyEnsemble**

RF	AUC	0.7804	0.7978	0.0174	16.37	0.0000
	F1-score	0.2681	0.3129	0.0448	55.88	0.0000
	MCC	0.2855	0.3242	0.0387	35.95	0.0000
	FP	558.0000	415.8556	-142.1444	-103.91	0.0000
	FN	14.9667	20.6444	5.6778	17.90	0.0000
XGBoost	AUC	0.7874	0.8199	0.0326	25.97	0.0000
	F1-score	0.2781	0.3496	0.0715	78.73	0.0000
	MCC	0.2960	0.3639	0.0679	50.06	0.0000
	FP	526.4445	354.5000	-171.9445	-216.62	0.0000
	FN	15.4556	19.2778	3.8222	11.79	0.0000

B.3. RandomUnderSampling

RF	AUC	0.7709	0.7878	0.0168	10.66	0.0000
	F1-score	0.2660	0.3082	0.0423	29.80	0.0000
	MCC	0.2776	0.3148	0.0372	20.71	0.0000
	FP	543.9778	411.6333	-132.3445	-46.24	0.0000
	FN	18.1667	23.2667	5.1000	13.61	0.0000
XGBoost	AUC	0.7543	0.7902	0.0359	19.94	0.0000
	F1-score	0.2544	0.3176	0.0632	42.61	0.0000
	MCC	0.2596	0.3236	0.0640	31.00	0.0000
	FP	556.2222	386.7222	-169.5000	-67.88	0.0000
	FN	21.3222	24.2222	2.9000	6.87	0.0000

Panel C. Forecasting Period is the Next Three Years

Model	Measure	Mean Before	Mean_After	Difference	T statistic	p-value
C.1. BalancedBagging						
RF	AUC	0.7940	0.7934	-0.0005	-0.74	0.4595
	F1-score	0.3216	0.3647	0.0431	56.64	0.0000
	MCC	0.3284	0.3592	0.0307	36.25	0.0000
	FP	575.8167	421.3500	-154.4667	-100.59	0.0000
	FN	35.2333	46.0833	10.8500	29.17	0.0000
XGBoost	AUC	0.7968	0.8033	0.0065	7.84	0.0000
	F1-score	0.3274	0.3938	0.0664	99.26	0.0000
	MCC	0.3344	0.3871	0.0526	63.10	0.0000
	FP	559.3000	365.5000	-193.8000	-174.56	0.0000
	FN	35.2500	46.2667	11.0167	28.34	0.0000
C.2. EasyEnsemble						
RF	AUC	0.7759	0.7919	0.0160	29.14	0.0000
	F1-score	0.2645	0.3092	0.0446	79.77	0.0000
	MCC	0.2808	0.3185	0.0378	66.67	0.0000
	FP	850.5667	627.3834	-223.1833	-112.07	0.0000
	FN	22.8833	32.4833	9.6000	27.53	0.0000
XGBoost	AUC	0.7794	0.8171	0.0377	63.23	0.0000
	F1-score	0.2719	0.3468	0.0749	97.40	0.0000
	MCC	0.2872	0.3601	0.0728	78.73	0.0000
	FP	806.8500	537.4833	-269.3667	-183.99	0.0000
	FN	24.6667	29.4500	4.7833	23.87	0.0000
C.3. RandomUnderSampling						
RF	AUC	0.7650	0.7805	0.0155	10.17	0.0000
	F1-score	0.2612	0.3024	0.0412	29.44	0.0000
	MCC	0.2710	0.3068	0.0358	20.16	0.0000

Table 9 (continued)**Panel A. Forecasting Period is the Next One Year**

	FP	832.4833	626.0667	−206.4166	−42.90	0.0000
	FN	28.0833	36.6500	8.5667	16.19	0.0000
XGBoost	AUC	0.7478	0.7831	0.0353	17.51	0.0000
	F1-score	0.2498	0.3111	0.0613	34.18	0.0000
	MCC	0.2527	0.3147	0.0619	25.70	0.0000
	FP	848.5333	590.9833	−257.5500	−56.32	0.0000
	FN	33.1667	38.0500	4.8833	7.50	0.0000

Panel D. Forecasting Period is the Next Four Years

Model	Measure	Mean Before	Mean_After	Difference	T statistic	p-value
D.1. BalancedBagging						
RF	AUC	0.7854	0.7820	−0.0033	−4.09	0.0003
	F1-score	0.3120	0.3528	0.0408	61.09	0.0000
	MCC	0.3175	0.3455	0.0280	30.10	0.0000
	FP	770.6000	559.1000	−211.5000	−193.30	0.0000
	FN	49.1333	64.9000	15.7667	42.57	0.0000
XGBoost	AUC	0.7885	0.7927	0.0042	5.29	0.0000
	F1-score	0.3179	0.3829	0.0650	82.51	0.0000
	MCC	0.3235	0.3743	0.0507	52.13	0.0000
	FP	748.4667	482.9333	−265.5334	−160.83	0.0000
	FN	49.1333	65.0000	15.8667	41.45	0.0000
D.2. EasyEnsemble						
RF	AUC	0.7730	0.7830	0.0100	16.08	0.0000
	F1-score	0.2602	0.3007	0.0405	89.56	0.0000
	MCC	0.2766	0.3083	0.0317	46.26	0.0000
	FP	1130.0670	832.5333	−297.5337	−174.77	0.0000
	FN	30.8333	46.1000	15.2667	51.70	0.0000
XGBoost	AUC	0.7716	0.8089	0.0372	46.97	0.0000
	F1-score	0.2637	0.3400	0.0763	154.28	0.0000
	MCC	0.2776	0.3511	0.0734	91.62	0.0000
	FP	1085.6330	705.0000	−380.6330	−238.34	0.0000
	FN	34.4333	42.5000	8.0667	20.26	0.0000
D.3. RandomUnderSampling						
RF	AUC	0.7594	0.7720	0.0126	7.21	0.0000
	F1-score	0.2551	0.2940	0.0389	22.06	0.0000
	MCC	0.2640	0.2968	0.0328	15.36	0.0000
	FP	1111.1330	832.4667	−278.6663	−28.02	0.0000
	FN	38.5000	51.2667	12.7667	17.72	0.0000
XGBoost	AUC	0.7387	0.7741	0.0354	14.05	0.0000
	F1-score	0.2411	0.3025	0.0614	27.90	0.0000
	MCC	0.2419	0.3040	0.0621	20.97	0.0000
	FP	1145.5000	784.1667	−361.3333	−43.82	0.0000
	FN	45.9000	53.5333	7.6333	7.38	0.0000

Note: AUC, F1-score, and MCC serve as the primary performance metrics for the model. Mean_Before and Mean_After represent the average model performance metrics before and after incorporating the EC_TCV variables, respectively (this study uses 30 sets of random states and a yearly rolling prediction method for each forecasting period in the bankruptcy prediction). The difference indicates the change in average performance metrics after incorporating EC_TCV variables, calculated as Mean_After minus Mean_Before. In addition, this study also applies various methods for handling data imbalance, including BalancedBagging, EasyEnsemble, and RandomUnderSampling

on prediction effectiveness of bankruptcy risk term structure are likely to increase with time, which is complementary to the theoretical expectation of Duffie and Lando (2001) that incomplete accounting information is more pronounced for short-term credit risk. The findings are consistent with the forward-looking characteristics of earnings call transcripts.

Table 9 also presents the results of the mean difference tests for the number of Type I and Type II errors in various bankruptcy prediction model settings. It has to be noted that Mean_Before represents the average performance of the model setting with Barboza et al.'s (2017) financial variables. Mean_After stands for the average performance of the model setting including earnings call transcripts *TCV* variables in addition to Barboza et al.'s (2017) financial variables. Empirical results of Table 9 show that the earnings call transcripts *TCV* variables have a significant decrease on the number of Type II error (FP) for all various model settings while have a slight increase on the number of Type I error (FN) for all various model settings for the future 1 to 4 years, especially for the XGBoost model with the EasyEnsemble and RandomUnderSampling algorithms. In addition, under the setting of the XGBoost model with the EasyEnsemble algorithm, the untabulated results present that earnings call transcripts *TCV* variables have the significant reductions on Type II error for the future 1 to 4 years by 0.0935, 0.0979, 0.1021, and 0.1088, whereas have the increases on Type I error for the future 1 to 4 years by 0.0403, 0.0328, 0.0267, and 0.0343, respectively.

Based on the above results, this study therefore concludes that the earnings call transcripts *TCV* variables provide the significantly incremental improvement in Type II error, especially for the longer forecasting periods. Meanwhile, the incremental increases on Type I error resulting from including earnings call transcripts *TCV* variables also decrease with longer forecasting periods.

4.2 Robustness results

To provide the more robust evidences for the importance of the earnings call transcripts *TCV* variables on bankruptcy prediction, this study (1) replaces the two employed machine learning models (RF, XGBoost) by another neural network model (e.g. deep neural network; DNN); (2) additionally introducing the Chen et al.'s (2023) 11 annual report *TCV* variables as control input variables. The two robustness results are listed as the following subsections:

4.2.1 Replacing the employed machine learning models by neural network models

To ensure the robustness of our main findings, this study also introduces neural network models (e.g. deep neural network) to replace the originally employed machine learning models for bankruptcy prediction analyses.⁷ For the deep neural network (DNN) model, this study follows Mai et al. (2019) and Chen et al. (2023) to employ DL-Deep model as the main model setting, such as four layers of hidden units in the network.⁸ This study also employs RandomOverSampling (ROS) and Synthesized Minority Oversampling Technique (SMOTE) algorithms to deal with data imbalance problem. Panel A (Panel B) in Table 10 shows the AUCs (F1 score) of these bankruptcy prediction models for future different periods (one to four years) using the DNN model. It has to be noted that *FIN* and *EC_TCV*

⁷ In addition to corporate bankruptcy prediction, Albanesi and Vamossy (2019) also discuss deep learning approaches for consumer default prediction.

⁸ Similar with Mai et al. (2019) and Chen et al. (2023), this study also uses Keras 2.11 package.

Table 10 Comparisons of the effectiveness of bankruptcy prediction with neural network models: financial variables vs. Earnings call transcripts TCV variables

Model	AUC in future 1 year	AUC in future 2 years	AUC in future 3 years	AUC in future 4 years
DNN&ROS				
FIN	0.7020	0.6898	0.6826	0.6695
FIN&EC_TCV	0.7537	0.7449	0.7447	0.7244
Difference	0.0517	0.0551	0.0620	0.0549
DNN&SMOTE				
FIN	0.6884	0.6774	0.6655	0.6564
FIN&EC_TCV	0.7358	0.7319	0.7223	0.7057
Difference	0.0474	0.0545	0.0568	0.0493
Model	F1-score in future 1 year	F1-score in future 2 years	F1-score in future 3 years	F1-score in future 4 years
DNN&ROS				
FIN	0.2209	0.2183	0.2090	0.1993
FIN&EC_TCV	0.2705	0.2664	0.2687	0.2575
Difference	0.0496	0.0481	0.0597	0.0582
DNN&SMOTE				
FIN	0.2143	0.2080	0.2007	0.1939
FIN&EC_TCV	0.3066	0.3086	0.3047	0.2885
Difference	0.0923	0.1006	0.1040	0.0946
Model	MCC in future 1 year	MCC in future 2 years	MCC in future 3 years	MCC in future 4 years
DNN&ROS				
FIN	0.1987	0.1911	0.1815	0.1689
FIN&EC_TCV	0.2677	0.2596	0.2615	0.2430
Difference	0.0691	0.0685	0.0800	0.0741
DNN&SMOTE				
FIN	0.1862	0.1766	0.1652	0.1565
FIN&EC_TCV	0.2820	0.2807	0.2726	0.2540
Difference	0.0958	0.1041	0.1075	0.0975

Note: Panel A (Panel B) in Table 10 shows the AUCs (F1 score) of these bankruptcy prediction models for future different periods, namely one to four years, using the deep neural network (DNN) model. *FIN* and *EC_TCV* present Barboza's et al. (2017) financial variables and our newly proposed earnings call transcripts *TCV* variables, respectively. This study also employs RandomOverSampling (ROS) and Synthesized Minority Oversampling Technique (SMOTE) algorithms to deal with data imbalance problem

present Barboza's et al. (2017) financial variables and our newly proposed earnings call transcripts *TCV* variables, respectively.

The empirical results in Table 10 indicate that incorporating earnings call transcripts' *TCV* features improves bankruptcy prediction effectiveness across all future forecasting periods.⁹ Notably, the incremental improvements in AUC (as well as F1 score and MCC) are more pronounced over longer periods (e.g., two or three years) than shorter ones (e.g., one year). These findings align with the forward-looking nature of earnings call transcripts.

The above results reveal that our main conclusions are robust for the DNN model setting. That is, our new engineered NLP feature variables, the earnings call transcripts *TCV* variables, can further improve the effectiveness of bankruptcy prediction in addition to Barboza et al.'s (2017) financial variables, especially for the long-term bankruptcy prediction.

⁹ The detailed descriptions about the improvement of bankruptcy prediction effectiveness could be referred to Appendix C.

4.2.2 Additionally introducing Chen_TCV variables as benchmark input variables

To ensure that earnings call transcripts *TCV* (*EC_TCV*) variables indeed provide particular information related to corporate bankruptcy that is distinct from annual report *TCV* variables (Chen et al. 2023), this study further introduces annual report *TCV* variables as additional benchmark input variables for bankruptcy prediction models. The detailed definitions of Chen et al.'s (2023) 11 annual report *TCV* variables (*Chen_TCV*) can be referred to Sect. 3.1.3 and Panel B of Table 1.

The empirical results in Tables 11, 12, and 13 show that incorporating earnings call transcripts *TCV* variables still substantially improves the bankruptcy prediction performance (measured by AUC, F1 score, and MCC) of machine learning models (RF, XGBoost) and neural network models (DNN), relative to models using Barboza_FIN and Chen_TCV variables alone. Unlike financial ratios and annual report textual data, which mainly capture historical outcomes, earnings call transcripts provide private, forward-looking, and timely information directly conveyed by management (Ning et al. 2025; Price et al. 2012; Fu et al. 2021). Moreover, earnings call transcripts incorporate analysts' questions and perspectives on the firm's future performance and uncertainties (e.g. Borochin et al. 2018), offering unique insights beyond those contained in traditional financial and textual measures. Therefore, the private, forward-looking, and timely nature of earnings call transcripts enables them to reflect managers' perspectives on future performance, risks, and uncertainties, thereby providing incremental predictive power in bankruptcy prediction beyond what retrospective financial ratios and annual report textual measures can offer.

Moreover, the results of Table 11, 12, and 13 all further indicate that the incremental benefits of additionally incorporating earnings call transcripts *TCV* variables are more pronounced over longer bankruptcy prediction periods. The detailed descriptions about empirical results are presented in Appendix D.

Accordingly, this study can conclude: (1) the information contents of earnings call transcripts *TCV* variables are complementary to those of annual report *TCV* variables (e.g. Chen et al. 2023) for debtholders; that is, earnings call transcripts *TCV* variables are complementary to the theoretical expectation of Duffie and Lando (2001) that incomplete accounting information is more pronounced for short-term credit risk; (2) the findings are robust for the assertion that earnings call transcripts *TCV* variables indeed provide particular information related to corporate bankruptcy that is distinct from annual report *TCV* variables (Chen et al. 2023), supporting the forward-looking characteristics of earnings call transcripts.

4.3 Interpretability analyses

To further explain which earnings call transcripts *TCV* variables primarily contribute to the effectiveness of bankruptcy prediction, this study employs the Random Forest and XGBoost methods to conduct feature engineering on Barboza et al.'s (2017) 11 financial (*FIN*) variables, Chen et al.'s (2023) 11 annual report *TCV* (*Chen_TCV*) variables, and the proposed 27 earnings call transcripts *TCV* (*EC_TCV*) variables over different training periods. The results are reported in Tables 14 and 15.¹⁰ The empirical evidence indicates that, among

¹⁰ In these two tables, it has to be noted that IMP_2016, IMP_2017, IMP_2018, and IMP_2019 represent the estimated relative importance of each feature using the training data set with the ending year of 2016, 2017, 2018, and 2019, respectively. IMP_Avg stands for the average of IMP_2016, IMP_2017, IMP_2018, and IMP_2019. This study employs IMP_Avg as the main ranking criteria.

Table 11 The AUCs comparisons of bankruptcy prediction models for future different periods: introducing annual report TCV variables as additional benchmark input variables

Model	AUC in future	AUC in future	AUC in future	AUC in future
	1 year	2 years	3 years	4 years
Panel A. Machine learning models				
<i>A.1. XGBoost&BalancedBagging</i>				
FIN&TCV	0.8280	0.8163	0.8088	0.8049
FIN&TCV&EC_TCV	0.8363	0.8278	0.8180	0.8128
Difference	0.0083	0.0115	0.0092	0.0079
<i>A.2. XGBoost&Easy Ensemble</i>				
FIN&TCV	0.8336	0.8203	0.8099	0.8024
FIN&TCV&EC_TCV	0.8393	0.8348	0.8263	0.8170
Difference	0.0057	0.0145	0.0164	0.0147
<i>A.3. XGBoost&RandomUnderSampling</i>				
FIN&TCV	0.8047	0.7962	0.7885	0.7823
FIN&TCV&EC_TCV	0.8184	0.8107	0.8028	0.7956
Difference	0.0137	0.0145	0.0144	0.0134
<i>A.4. RF&BalancedBagging</i>				
FIN&TCV	0.8211	0.8121	0.8034	0.8015
FIN&TCV&EC_TCV	0.8223	0.8159	0.8085	0.8029
Difference	0.0012	0.0039	0.0050	0.0014
<i>A.5. RF&Easy Ensemble</i>				
FIN&TCV	0.8096	0.8000	0.7939	0.7889
FIN&TCV&EC_TCV	0.8132	0.8089	0.8022	0.7946
Difference	0.0035	0.0089	0.0083	0.0057
<i>A.6. RF&RandomUnderSampling</i>				
FIN&TCV	0.8050	0.7952	0.7883	0.7850
FIN&TCV&EC_TCV	0.8057	0.7993	0.7922	0.7855
Difference	0.0007	0.0041	0.0039	0.0005
Panel B. Deep neural network (DNN)				
<i>B.1. DNN&ROS</i>				
FIN&TCV	0.7664	0.7508	0.7420	0.7308
FIN&TCV&EC_TCV	0.8043	0.7970	0.7819	0.7761
Difference	0.0379	0.0461	0.0399	0.0453
<i>B.2. DNN&SMOTE</i>				
FIN&TCV	0.7381	0.7169	0.7080	0.7013
FIN&TCV&EC_TCV	0.7666	0.7561	0.7439	0.7250
Difference	0.0285	0.0392	0.0359	0.0237

Note: Panel A and Panel B in Table 11 present the comparative analyses for the prediction effectiveness (measured by AUC) of machine learning models (e.g. RF and XGBoost) and neural network models (e.g. DNN) using two categories of input variables, respectively. One group consists of *Barboza_FIN* and *Chen_TCV* variables, while the other group includes the *Barboza_FIN*, *Chen_TCV*, and earnings call transcripts *TCV (EC_TCV)* variables. The former group is called as FIN&TCV, and the latter group is called as FIN&TCV&EC_TCV

the *EC_TCV* variables, the positive tone (Full_POS) and uncertainty tone (Full_UNC) emerge as the two most influential factors.¹¹ For these two tone variables, Full_POS captures the degree of optimistic expectations, whereas Full_UNC reflects the precision of the firm's forward-looking communicative value regarding its future asset value distributions.

¹¹ The average of the estimated relative importance of Full_POS and Full_UNC variables under the XGBoost (Random Forest) feature engineering approaches are 0.0286 and 0.0320 (0.0255 and 0.0254), respectively. Besides, these two features rank as the fourth to fifth (ninth to tenth) most important among all 49 feature variables under the XGBoost (Random Forest) feature engineering approaches.

Table 12 The F1 score comparisons of bankruptcy prediction models for future different periods: introducing annual report TCV variables as additional benchmark input variables

Model	F1-score in future	F1-score in future	F1-score in future	F1-score in future
	1 year	2 years	3 years	4 years
Panel A. Machine learning models				
<i>A.1. XGBoost&BalancedBagging</i>				
FIN&TCV	0.3839	0.3724	0.3610	0.3495
FIN&TCV&EC_TCV	0.4196	0.4119	0.4018	0.3938
Difference	0.0357	0.0394	0.0408	0.0443
<i>A.2. XGBoost&Easy Ensemble</i>				
FIN&TCV	0.3371	0.3280	0.3175	0.3036
FIN&TCV&EC_TCV	0.3731	0.3707	0.3615	0.3485
Difference	0.0360	0.0427	0.0440	0.0449
<i>A.3. XGBoost&RandomUnderSampling</i>				
FIN&TCV	0.3116	0.3052	0.2970	0.2860
FIN&TCV&EC_TCV	0.3459	0.3404	0.3326	0.3214
Difference	0.0343	0.0353	0.0357	0.0354
<i>A.4. RF&BalancedBagging</i>				
FIN&TCV	0.3693	0.3624	0.3522	0.3455
FIN&TCV&EC_TCV	0.3920	0.3882	0.3802	0.3727
Difference	0.0227	0.0258	0.0280	0.0272
<i>A.5. RF&Easy Ensemble</i>				
FIN&TCV	0.3103	0.3036	0.2968	0.2880
FIN&TCV&EC_TCV	0.3323	0.3293	0.3234	0.3137
Difference	0.0220	0.0257	0.0266	0.0257
<i>A.6. RF&RandomUnderSampling</i>				
FIN&TCV	0.3089	0.3022	0.2953	0.2880
FIN&TCV&EC_TCV	0.3257	0.3220	0.3155	0.3072
Difference	0.0168	0.0198	0.0202	0.0192
Panel B. Deep neural network (DNN)				
<i>B.1. DNN&ROS</i>				
FIN&TCV	0.3244	0.3103	0.3027	0.2797
FIN&TCV&EC_TCV	0.3830	0.3823	0.3641	0.3548
Difference	0.0585	0.0720	0.0615	0.0751
<i>B.2. DNN&SMOTE</i>				
FIN&TCV	0.2949	0.2804	0.2724	0.2639
FIN&TCV&EC_TCV	0.3771	0.3700	0.3626	0.3459
Difference	0.0822	0.0897	0.0902	0.0821

Panel A and Panel B in Table 12 present the comparative analyses for the prediction effectiveness (measured by F1 score) of machine learning models (e.g. RF and XGBoost) and neural network models (e.g. DNN) using two categories of input variables, respectively. One group consists of *Barboza_FIN* and *Chen_TCV* variables, while the other group includes the *Barboza_FIN*, *Chen_TCV*, and earnings call transcripts *TCV* (*EC_TCV*) variables. The former group is called as FIN&TCV, and the latter group is called as FIN&TCV&EC_TCV

Together, these variables not only characterize the firm's future asset value distributions but also signal its prospective financial condition, thereby reflecting corporate bankruptcy risk. In addition, the more forward-looking information provided by earnings call transcripts is consistent with a more noticeable improvement effect on the effectiveness of long-term bankruptcy prediction, which is complementary to the views of Duffie and Lando (2001) and Chen et al. (2023), who emphasize the substantial impact of incomplete accounting information on short-term credit risk assessment.

Table 13 The F1 score comparisons of bankruptcy prediction models for future different periods: introducing annual report TCV variables as additional benchmark input variables

Model	F1- score in future	F1- score in future	F1- score in future	F1- score in future
	1 year	2 years	3 years	4 years
Panel A. Machine learning models				
<i>A.1. XGBoost&BalancedBagging</i>				
FIN&TCV	0.3839	0.3724	0.3610	0.3495
FIN&TCV&EC_TCV	0.4196	0.4119	0.4018	0.3938
Difference	0.0357	0.0394	0.0408	0.0443
<i>A.2. XGBoost&Easy Ensemble</i>				
FIN&TCV	0.3371	0.3280	0.3175	0.3036
FIN&TCV&EC_TCV	0.3731	0.3707	0.3615	0.3485
Difference	0.0360	0.0427	0.0440	0.0449
<i>A.3. XGBoost&RandomUnderSampling</i>				
FIN&TCV	0.3116	0.3052	0.2970	0.2860
FIN&TCV&EC_TCV	0.3459	0.3404	0.3326	0.3214
Difference	0.0343	0.0353	0.0357	0.0354
<i>A.4. RF&BalancedBagging</i>				
FIN&TCV	0.3693	0.3624	0.3522	0.3455
FIN&TCV&EC_TCV	0.3920	0.3882	0.3802	0.3727
Difference	0.0227	0.0258	0.0280	0.0272
<i>A.5. RF&Easy Ensemble</i>				
FIN&TCV	0.3103	0.3036	0.2968	0.2880
FIN&TCV&EC_TCV	0.3323	0.3293	0.3234	0.3137
Difference	0.0220	0.0257	0.0266	0.0257
<i>A.6. RF&RandomUnderSampling</i>				
FIN&TCV	0.3089	0.3022	0.2953	0.2880
FIN&TCV&EC_TCV	0.3257	0.3220	0.3155	0.3072
Difference	0.0168	0.0198	0.0202	0.0192
Panel B. Deep neural network (DNN)				
<i>B.1. DNN&ROS</i>				
FIN&TCV	0.3244	0.3103	0.3027	0.2797
FIN&TCV&EC_TCV	0.3830	0.3823	0.3641	0.3548
Difference	0.0585	0.0720	0.0615	0.0751
<i>B.2. DNN&SMOTE</i>				
FIN&TCV	0.2949	0.2804	0.2724	0.2639
FIN&TCV&EC_TCV	0.3771	0.3700	0.3626	0.3459
Difference	0.0822	0.0897	0.0902	0.0821

Panel A and Panel B in Table 12 present the comparative analyses for the prediction effectiveness (measured by F1 score) of machine learning models (e.g. RF and XGBoost) and neural network models (e.g. DNN) using two categories of input variables, respectively. One group consists of *Barboza_FIN* and *Chen_TCV* variables, while the other group includes the *Barboza_FIN*, *Chen_TCV*, and earnings call transcripts *TCV* (*EC_TCV*) variables. The former group is called as FIN&TCV, and the latter group is called as FIN&TCV&EC_TCV

4.4 Interpretability analyses

Furthermore, we also further discuss why positive tone (Full_POS) and uncertainty tone (Full_UNC) emerge as the most influential earnings call transcripts *TCV* variables in predicting corporate bankruptcy from the theoretical perspectives of structural-form credit risk models (e.g., Merton 1974; Duffie and Lando 2001). Specifically, the use of positive tone

Table 14 Ranking of importance of all input variables used in bankruptcy prediction models (Each training Period): random forest

Rank	Group	Feature	IMP 2016	IMP 2017	IMP 2018	IMP 2019	IMP_Avg
1	(A). Barboza's (2017) FIN	EBIT_Sales	0.0430	0.0410	0.0399	0.0390	0.0408
2	(A). Barboza's (2017) FIN	EBIT_TA	0.0396	0.0412	0.0399	0.0400	0.0402
3	(A). Barboza's (2017) FIN	EMV_Debt	0.0403	0.0392	0.0383	0.0393	0.0393
4	(A). Barboza's (2017) FIN	Sales_TA	0.0309	0.0313	0.0315	0.0319	0.0314
5	(A). Barboza's (2017) FIN	NWC_TA	0.0261	0.0358	0.0294	0.0337	0.0312
6	(B). Chen's (2023) TCV	FT_POS	0.0275	0.0266	0.0308	0.0301	0.0287
7	(B). Chen's (2023) TCV	FSIZE	0.0286	0.0269	0.0267	0.0251	0.0268
8	(A). Barboza's (2017) FIN	TA_growth	0.0271	0.0271	0.0251	0.0234	0.0257
9	(C). EC_TCV	Full_POS	0.0250	0.0237	0.0285	0.0248	0.0255
10	(C). EC_TCV	Full_UNC	0.0226	0.0273	0.0269	0.0249	0.0254
11	(C). EC_TCV	Full_Mweak	0.0224	0.0232	0.0274	0.0261	0.0248
12	(B). Chen's (2023) TCV	A_WPP	0.0252	0.0218	0.0247	0.0228	0.0236
13	(B). Chen's (2023) TCV	Harv_NEG	0.0217	0.0231	0.0256	0.0235	0.0235
14	(A). Barboza's (2017) FIN	EMP_growth	0.0273	0.0214	0.0212	0.0239	0.0235
15	(A). Barboza's (2017) FIN	RE_TA	0.0219	0.0241	0.0225	0.0242	0.0232
16	(A). Barboza's (2017) FIN	ROE_Chg	0.0257	0.0221	0.0229	0.0214	0.0230
17	(C). EC_TCV	Full_Constr	0.0233	0.0235	0.0196	0.0240	0.0226
18	(C). EC_TCV	Full_WPS	0.0233	0.0228	0.0216	0.0223	0.0225
19	(B). Chen's (2023) TCV	FK_Ease	0.0220	0.0223	0.0223	0.0224	0.0223
20	(B). Chen's (2023) TCV	FT_UNC	0.0224	0.0217	0.0209	0.0215	0.0216
21	(B). Chen's (2023) TCV	FT_MWeak_C	0.0198	0.0209	0.0211	0.0230	0.0212
22	(B). Chen's (2023) TCV	CL	0.0201	0.0184	0.0225	0.0224	0.0209
23	(B). Chen's (2023) TCV	FT_NEG	0.0214	0.0186	0.0200	0.0210	0.0203
24	(A). Barboza's (2017) FIN	Sales_growth	0.0173	0.0188	0.0209	0.0221	0.0198
25	(B). Chen's (2023) TCV	FT_MStr	0.0183	0.0183	0.0167	0.0200	0.0183
26	(C). EC_TCV	Full_NEG	0.0199	0.0162	0.0169	0.0186	0.0179
27	(C). EC_TCV	Full_MStr	0.0191	0.0175	0.0169	0.0157	0.0173
28	(A). Barboza's (2017) FIN	PB_Chg	0.0174	0.0179	0.0176	0.0149	0.0169
29	(C). EC_TCV	Full_ARI	0.0167	0.0141	0.0184	0.0170	0.0165
30	(C). EC_TCV	Full_Complex_words	0.0191	0.0155	0.0144	0.0145	0.0159
31	(B). Chen's (2023) TCV	FT_LITI	0.0160	0.0164	0.0159	0.0148	0.0158
32	(C). EC_TCV	Full_FOG	0.0152	0.0161	0.0149	0.0163	0.0156
33	(C). EC_TCV	Full_LITI	0.0164	0.0168	0.0142	0.0148	0.0156
34	(C). EC_TCV	Full_CL	0.0159	0.0156	0.0153	0.0154	0.0155
35	(C). EC_TCV	Full_Char_PW	0.0138	0.0150	0.0165	0.0153	0.0151
36	(C). EC_TCV	Full_Syll_PW	0.0155	0.0155	0.0151	0.0144	0.0151

Note: The above variables are top 36 important feature using the Random Forest algorithm from the research variables of Barboza et al.'s (2017) 11 financial (FIN) variables, 11 Chen et al.'s (2023) TCV variables, and the newly proposed 27 EC_TCV variables in this study. The detailed definitions of FIN, Chen_TCV, and EC_TCV variables can be referred to those in Tables 1 and 2, and 3, respectively. In addition, IMP_2016, IMP_2017, IMP_2018, and IMP_2019 represent the estimated relative importance of each feature using the training data set with the ending year of 2016, 2017, 2018, and 2019, respectively. IMP_Avg stands for the average of IMP_2016, IMP_2017, IMP_2018, and IMP_2019

Table 15 Ranking of importance of all input variables used in bankruptcy prediction models (Each training Period): XGBoost

Rank	Group	Feature	IMP 2016	IMP 2017	IMP 2018	IMP 2019	IMP_Avg
1	(A). Barboza's (2017) FIN	EBIT_TA	0.0544	0.0575	0.0484	0.0668	0.0568
2	(C). EC_TCV	Full_Lwords	0.0663	0.0342	0.0380	0.0338	0.0431
3	(A). Barboza's (2017) FIN	EBIT_Sales	0.0312	0.0326	0.0319	0.0379	0.0334
4	(C). EC_TCV	Full_UNC	0.0312	0.0362	0.0299	0.0309	0.0320
5	(C). EC_TCV	Full_POS	0.0291	0.0333	0.0240	0.0280	0.0286
6	(B). Chen's (2023) TCV	FK_Ease	0.0289	0.0327	0.0266	0.0191	0.0268
7	(A). Barboza's (2017) FIN	EMV_Debt	0.0227	0.0264	0.0270	0.0282	0.0261
8	(B). Chen's (2023) TCV	FT_MWeak_C	0.0231	0.0278	0.0254	0.0228	0.0248
9	(B). Chen's (2023) TCV	FT_POS	0.0250	0.0335	0.0206	0.0191	0.0245
10	(A). Barboza's (2017) FIN	RE_TA	0.0207	0.0259	0.0243	0.0236	0.0236
11	(B). Chen's (2023) TCV	FSIZE	0.0234	0.0233	0.0246	0.0228	0.0235
12	(A). Barboza's (2017) FIN	NWC_TA	0.0251	0.0278	0.0182	0.0217	0.0232
13	(B). Chen's (2023) TCV	FT_UNC	0.0237	0.0276	0.0203	0.0202	0.0230
14	(A). Barboza's (2017) FIN	Sales_TA	0.0230	0.0210	0.0232	0.0236	0.0227
15	(C). EC_TCV	Full_Words	0.0084	0.0111	0.0440	0.0251	0.0221
16	(A). Barboza's (2017) FIN	EMP_growth	0.0220	0.0232	0.0225	0.0197	0.0218
17	(C). EC_TCV	Full_SMOG	0.0132	0.0211	0.0221	0.0301	0.0216
18	(C). EC_TCV	Full_MStr	0.0323	0.0161	0.0227	0.0148	0.0215
19	(B). Chen's (2023) TCV	CL	0.0213	0.0187	0.0195	0.0263	0.0214
20	(B). Chen's (2023) TCV	Harv_NEG	0.0264	0.0198	0.0178	0.0211	0.0213
21	(B). Chen's (2023) TCV	FT_NEG	0.0189	0.0199	0.0257	0.0202	0.0212
22	(C). EC_TCV	Full_CL	0.0167	0.0214	0.0309	0.0148	0.0210
23	(C). EC_TCV	Full_WPS	0.0174	0.0241	0.0210	0.0195	0.0205
24	(C). EC_TCV	Full_Mweak	0.0193	0.0264	0.0193	0.0169	0.0205
25	(B). Chen's (2023) TCV	A_WPP	0.0197	0.0178	0.0169	0.0231	0.0194
26	(A). Barboza's (2017) FIN	Sales_growth	0.0201	0.0200	0.0177	0.0189	0.0192
27	(C). EC_TCV	Full_Dale	0.0171	0.0167	0.0207	0.0205	0.0188
28	(A). Barboza's (2017) FIN	ROE_Chg	0.0195	0.0174	0.0182	0.0190	0.0185
29	(C). EC_TCV	Full_Chars	0.0092	0.0137	0.0068	0.0437	0.0184
30	(C). EC_TCV	Full_ARI	0.0133	0.0129	0.0185	0.0277	0.0181
31	(A). Barboza's (2017) FIN	TA_growth	0.0189	0.0148	0.0199	0.0186	0.0181
32	(C). EC_TCV	Full_LITI	0.0176	0.0189	0.0145	0.0200	0.0178
33	(C). EC_TCV	Full_Syll_PW	0.0158	0.0108	0.0288	0.0153	0.0176
34	(C). EC_TCV	Full_Constr	0.0199	0.0189	0.0157	0.0158	0.0176
35	(C). EC_TCV	Full_FOG	0.0127	0.0177	0.0251	0.0106	0.0165
36	(B). Chen's (2023) TCV	FT_LITI	0.0189	0.0119	0.0196	0.0145	0.0162
37	(A). Barboza's (2017) FIN	PB_Chg	0.0154	0.0212	0.0109	0.0159	0.0159
38	(C). EC_TCV	Full_Char_PW	0.0155	0.0179	0.0122	0.0158	0.0153
39	(C). EC_TCV	Full_Sylls	0.0128	0.0191	0.0160	0.0125	0.0151

Note: The above variables are top 36 important feature using the XGBoost algorithm from the research variables of Barboza et al.'s (2017) 11 financial (FIN) variables, 11 Chen et al.'s (2023) TCV variables, and the newly proposed 27 EC_TCV variables in this study. The detailed definitions of FIN, Chen_TCV, and EC_TCV variables can be referred to those in Tables 1 and 2, and 3, respectively. In addition, IMP_2016, IMP_2017, IMP_2018, and IMP_2019 represent the estimated relative importance of each feature using the training data set with the ending year of 2016, 2017, 2018, and 2019, respectively. IMP_Avg stands for the average of IMP_2016, IMP_2017, IMP_2018, and IMP_2019

can signal future firm performance (Davis et al. 2012), reduce uncertainty about firm value (Borochin et al. 2018), and mitigate investors' concerns regarding information asymmetry, thereby enhancing market confidence in the firm (Davis et al. 2012; Ning et al. 2025). The uncertainty tone of earnings call transcripts implies that managers may currently be exposed to operational risks arising from managerial or environmental pressures. Since these two tone variables directly capture several theoretical determinants in structural-form credit risk models (e.g., incomplete information, asset value volatility, and asset value), they are likely to be more sensitive in predicting bankruptcy risk than other textual features, whose effects on these core determinants are relatively mixed or less evident. The detailed discussions are provided in Appendix E.

In addition, to further analyze the impact direction of the selected *EC_TCV* variables on the effectiveness of the bankruptcy prediction model, this study employs SHAP analysis (Lundberg and Lee 2017) to enhance the model's interpretability. From the results of SHAP analysis, earnings call transcripts positive tone (Full_POS) variable and earnings call transcripts uncertainty tone (Full_UNC) variable consistently exhibits relatively high and stable average absolute SHAP values, ranking the top six for the next one- to four-year bankruptcy predictions. Moreover, the study also finds that Full_POS variable is negatively related to corporate bankruptcy risk while Full_UNC variable has the opposite effect. The SHAP Summary Plots (Figures 1a to 4a), SHAP Feature Importance (Figures 1b to 4b), and related discussions are presented in Appendix E.

5 Conclusions

This study primarily aims to explore whether including earnings call transcripts *TCV* variables improve the effectiveness of bankruptcy prediction models utilizing machine learning and data imbalance processing algorithms. Also, the study aims to provide additional insights and implications for bank loan practices. In the literature that employs machine learning models for bankruptcy prediction, the majority focuses on a firm's financial variables and the extent of textual information in annual reports. However, there is limited attention given to incorporating textual information from earnings call transcripts. Our results indicate that including the earnings call transcripts *TCV* variables not only significantly increase the F1 scores, AUCs, and MCCs of bankruptcy prediction models but also leads to a great decrease on the number of Type II error under the little change or a slight increase on the number of Type I error compared to Barboza et al.'s (2017) model. The results suggest that the earnings call transcripts *TCV* variables can significantly reduce the number of misjudging non-bankrupt firms as bankrupt ones, namely decrease the opportunity for banks not to grant credit loans to customers that actually own sound solvency profiles. Hence, earnings call transcripts *TCV* variables can effectively help banks in their decision-making of granting credit loans, and further enhance the efficiency of capital utilization, strengthen the operating performance of credit granting business, and ultimately reduce the probability of credit contagion in the supply chain.

In addition, this study further finds that the incremental impacts of the earnings call transcripts *TCV* variables on prediction effectiveness of bankruptcy risk term structure are likely to increase more in longer-term forecasting periods, which is complementary to the theoretical expectation of Duffie and Lando (2001) that incomplete accounting informa-

tion is more pronounced for short-term credit risk. These findings are consistent with the forward-looking characteristics of earnings call transcripts. Moreover, the above results are robust for the neural network model settings and introducing annual report *TCV* variables as additional benchmark input variables for bankruptcy prediction models. Furthermore, the model with the earnings call transcripts *TCV* variables have such a significant improvement in bankruptcy prediction compared to Barboza et al.'s (2017) model and Chen et al.'s (2023) model, which also verifies the assertion that earnings call transcripts *TCV* variables indeed provide particular information related to corporate bankruptcy that is distinct from annual report *TCV* variables (Chen et al. 2023), supporting the forward-looking characteristics of earnings call transcripts. Therefore, this study can conclude that the earnings call transcripts *TCV* variables indeed provide additional explanatory power for corporate bankruptcy predictions. Finally, this study further provides evidences that Full_POS and Full_UNC variables serve as the most crucial bankruptcy prediction indicators among earnings call transcripts *TCV* variables. These two features respectively reflect the degree of optimistic expectations about a firm's future asset value distributions and the precision level of the communicative value of the firm's forward-looking information. Together, they not only signal the firm's prospective financial condition but also characterize its future asset value distributions, which, in turn, shape corporate bankruptcy risk in line with structural-form credit risk models (Merton 1974; Duffie and Lando 2001). Consistent with theoretical expectations, the SHAP analysis results indicate that the Full_POS variable is negatively associated with corporate bankruptcy risk, whereas the Full_UNC variable exhibits a positive association.

The main contributions of this study include: (1) demonstrating the economic implications and importance of the earnings call transcripts *TCV* variables into bankruptcy predictions models; (2) proposing the newly engineered NLP features of the earnings call transcripts *TCV* variables, which are different from other mentioned variables in annual report textual literature (Mai et al. 2019; Chen et al. 2023), such as annual report readability and tones, and word frequency and sentiment in the MD&A sections; (3) providing the solid theoretical foundations of the earnings call transcripts *TCV* variables on bankruptcy prediction models; (4) exploring that the incremental contribution of earnings call transcripts *TCV* variables becomes more pronounced in longer-term bankruptcy predictions, which aligns with the forward-looking nature of earnings call transcripts and also complements the findings of Chen et al. (2023) regarding annual report *TCV* variables; (5) indicating that Full_POS and Full_UNC are the most critical earnings call transcripts *TCV* variables for predicting corporate bankruptcy, with Full_POS negatively and Full_UNC positively associated with bankruptcy risk; (6) providing the practical suggestions for banks to further facilitate more efficient capital allocation, enhance the operating performance of credit granting business, and reduce the probability of credit contagion in the supply chain since earnings call transcripts *TCV* variables significantly decrease the occurrence of misjudging non-bankrupt firms as bankrupt ones; (7) verifying that earnings call transcripts *TCV* variables indeed provide specific bankruptcy-related information that is different from annual report *TCV* variables (Chen et al. 2023), supporting the forward-looking characteristics of earnings call transcripts. Finally, for financial institutions and regulatory authorities, leveraging the textual information about the earnings call transcripts can aid in assessing whether a firm has bankruptcy concerns in the future (especially for the longer-term forecasting periods) and facilitating informed lending decisions or regulatory measures.

A key limitation of this study is the reliance on dictionary-based methods (e.g., Loughran and McDonald 2011), which, while widely applied in financial text analysis and valued for their transparency and interpretability, remain rigid and context-insensitive. These methods often struggle with negation, evolving language, and contextual tone, which may constrain their effectiveness in capturing nuanced managerial sentiment. Future research could explore more advanced NLP approaches, such as context-aware models (e.g. Vamossoy and Skog 2025) or large language models (LLMs), which are better equipped to interpret linguistic subtleties and may improve the accuracy of bankruptcy prediction.

Appendix

Appendix A. The details of the text-based communicative value variables of earnings call transcript

This study builds upon Chen et al. (2023) to develop 27 text-based communicative value (*TCV*) variables for earnings call transcripts using the readability suite in Python and the Loughran-McDonald dictionary (Loughran and McDonald 2011, 2014). The newly proposed 27 earnings call transcripts *TCV* (*EC_TCV*) variables capture three complementary dimensions of textual characteristics—readability, text structure complexity, and tone or sentiment—which together reflect how effectively management communicates information during earnings call conferences. First, the readability dimension comprises nine classic linguistic indices that measure the ease of text comprehension, including: (1) Flesch–Kincaid Grade Level (Flesch 1948), (2) Automated Readability Index (Smith and Senter 1967), (3) Coleman–Liau Index (Coleman and Liau 1975), (4) Flesch Reading Ease (Flesch 1948), (5) Gunning Fog Index (Gunning 1952), (6) LIX Index (Björnsson, 1968), (7) SMOG Grade (McLaughlin 1969), (8) RIX Index (Anderson, 1983), and (9) Dale–Chall Score (Dale and Chall 1948). The above nine readability variables of earnings call transcripts are denoted as Full_FK, Full_ARI, Full_CL, Full_FK_Ease, Full_FOG, Full_LIX, Full_SMOG, Full_RIX, and Full_Dale, respectively.

Second, the text structure complexity dimension includes eleven variables that describe the quantitative and syntactic structure of transcripts, such as: (1) average number of characters per word, (2) average syllables per word, (3) average words per sentence, (4) total characters, (5) total number of words (python-readability package based), (6) total sentences, (7) total long words, (8) total complex words, (9) total number of words (blank based), (10) total words appearing in the Loughran–McDonald master dictionary, and (11) total syllables. The above eleven text structure complexity variables of earnings call transcripts are labeled as Full_Char_PW, Full_Syll_PW, Full_WPS, Full_Chars, Full_Words, Full_Sents, Full_Lwords, Full_Complex_words, Full_Twords, Full_countedinMcD, and Full_Sylls, respectively.

Finally, the tone and sentiment dimension consists of seven variables derived from the Loughran–McDonald dictionary, capturing the emotional and modal aspects of managerial communication: (1) negative tone, (2) positive tone, (3) uncertainty tone, (4) litigious tone, (5) financial-modal strong tone, (6) financial-modal weak tone, and (7) constraining tone. The above seven tone and sentiment variables of earnings call transcripts are presented as Full_NEG, Full_POS, Full_UNC, Full_LITI, Full_MStr, Full_MWeak, and Full_Constr, respectively.

Appendix B. The details of imbalanced data processing methods

To mitigate potential biases arising from class imbalance, this study applies three methods—EasyEnsemble, BalancedBaggingClassifier, and Random Undersampling. The methodological details are described below.

Regarding the EasyEnsemble algorithm, it is a combined learning approach that integrates Bagging and Adaboost methods. Its core concept involves the following steps: (1) conducting random sampling k times from the majority category sample; (2) extracting an equivalent number of samples from the minority category sample in each sampling round, resulting in the creation of k datasets; (3) training these datasets k times, resulting in the generation of k distinct models; (4) employing a majority vote mechanism to derive the final outcome. Furthermore, this study utilizes the EasyEnsemble algorithm to generate 1000 subsets through 1000 random samplings in the implementation. Subsequently, 1000 sub-models are created, and a majority decision is applied to determine the final result.

In relation to the BalanceBaggingClassifier algorithm, it is a hybrid approach that combines a base classifier with the EasyEnsemble algorithm. It is important to highlight that the base classifier can be specified using the parameter setting “base estimator”. Then the final classification results are obtained by amalgamating outputs from multiple models. In addition, the concept of Random Undersampling revolves around randomly removing samples from the majority class. This process aims to achieve class balance by ensuring that all classes have an equal quantity of instances.

Appendix C. Detailed discussions of the empirical results using the deep neural network model

Table 10 presents that the AUCs, F1 score and MCC of the bankruptcy prediction using deep neural network (DNN) model for each future forecasting period have been also improved after adding earnings call transcripts *TCV* features. For instance, (1) the AUCs of the bankruptcy prediction in the future one (two, three, four) year for the DNN model with ROS algorithm can rise from 0.7020 to 0.7537 (0.6898 to 0.7449, 0.6826 to 0.7447, 0.6695 to 0.7244) after adding earnings call transcripts *TCV* features; (2) the F1 score of the bankruptcy prediction in the future one (two, three, four) year for the DNN model with SMOTE algorithm can rise from 0.2143 to 0.3066 (0.2080 to 0.3086, 0.2007 to 0.3047, 0.1939 to 0.2885) after adding earnings call transcripts *TCV* features; (3) the MCC of the bankruptcy prediction in the future one (two, three, four) year for the DNN model with SMOTE algorithm can rise from 0.1862 to 0.2820 (0.1766 to 0.2807, 0.1652 to 0.2726, 0.1565 to 0.2540) after adding earnings call transcripts *TCV* features. The incremental AUCs (F1 score, MCC) of the bankruptcy prediction models after including earnings call transcripts *TCV* features are higher in longer time periods (e.g. two or three years) compared with the shorter time period (e.g. one year). The findings are consistent with the forward-looking characteristics of earnings call transcripts.

Appendix D. Detailed descriptions of the empirical results with the inclusion of Chen_TCV variables as additional benchmark inputs

Panel A and Panel B in Tables 11, 12, and 13 present the comparative analyses for the prediction effectiveness (measured by AUC, F1 score, and MCC) of machine learning models

(e.g. RF and XGBoost) and neural network models (e.g. DNN) using two categories of input variables, respectively. One group consists of *Barboza_FIN* and *Chen_TCV* variables, while the other group includes the *Barboza_FIN*, *Chen_TCV*, and earnings call transcripts *TCV* (*EC_TCV*) variables. The former group is called as FIN&TCV, and the latter group is called as FIN&TCV&EC_TCV.

The empirical findings in Panel A of Tables 11, 12, and 13 indicate that, regardless of BalancedBagging, EasyEnsemble, and RandomUnderSampling algorithms, the additional inclusion of the *EC_TCV* variables substantially improves the bankruptcy prediction effectiveness of machine learning models (RF, XGBoost) compared to the models using *Barboza_FIN* and *Chen_TCV* variables as model input variables. For instance, when additionally incorporating earnings call transcripts *TCV* variables into the XGBoost model with EasyEnsemble (BalancedBagging) algorithm, the AUC (F1 score; MCC) for the next year increases from 0.8336 to 0.8393 (0.3839 to 0.4196; 0.3936 to 0.4248). Similarly, the AUC (F1 score; MCC) for the next two years increases from 0.8203 to 0.8348 (0.3724 to 0.4119; 0.3779 to 0.4130), and for the next three years, it increases from 0.8099 to 0.8263 (0.3610 to 0.4018; 0.3654 to 0.4008). Additionally, the AUC (F1 score; MCC) for the next four years also experiences an increase, growing from 0.8024 to 0.8170 (0.3495 to 0.3938; 0.3556 to 0.3930). The results suggest that the AUC (F1 score; MCC) increases in the next one, two, three and four years under this model setting are 0.0057, 0.0145, 0.0164 and 0.0146 (0.0357, 0.0395, 0.0408, and 0.0443; 0.0312, 0.0351, 0.0354, and 0.0374), respectively.

Similarly, the empirical findings in Panel B of Tables 11, 12, and 13 indicate that, regardless of ROS and SMOTE algorithms, the additional inclusion of the *EC_TCV* variables also considerably improves the bankruptcy prediction effectiveness of neural network models (DNN) compared to the models using *Barboza_FIN* and *Chen_TCV* variables as model input variables. For instance, when additionally incorporating earnings call transcripts *TCV* variables into the DNN model with ROS algorithm, the AUC (F1 score; MCC) for the next year increases from 0.7664 to 0.8043 (0.3244 to 0.3830; 0.3079 to 0.3775). Similarly, the AUC for the next two years increases from 0.7508 to 0.7970 (0.3103 to 0.3823; 0.2908 to 0.3721), and for the next three years, it increases from 0.7420 to 0.7819 (0.3027 to 0.3641; 0.2816 to 0.3509). Additionally, the AUC for the next four years also experiences an increase, growing from 0.7308 to 0.7761 (0.2797 to 0.3548; 0.2599 to 0.3428). The results suggest that the AUC (F1 score; MCC) increases in the next one, two, three and four years under this model setting are 0.0379, 0.0462, 0.0399 and 0.0453 (0.0586, 0.0720, 0.0614, and 0.0751; 0.0697, 0.0813, 0.0692, and 0.0829), respectively.

Appendix E. Further discussions for feature engineering interpretation

This section explains why positive tone (Full_POS) and uncertainty tone (Full_UNC) of earnings call transcripts emerge as the most influential earnings call transcripts *TCV* variables in predicting corporate bankruptcy, as well as discussing their impacts on bankruptcy outcomes. We address this issue from the theoretical perspective of structural-form credit risk models. Within this framework (e.g., Merton 1974; Duffie and Lando 2001), four core determinants of firm credit risk are identified: asset value, asset value volatility, the default threshold, and incomplete information. Among these, asset value is negatively associated with firm credit risk, whereas the other three components exhibit a positive relationship.

The possible explanations for why positive and uncertainty tone variables of earnings call transcripts emerge as the most influential predictors among earnings call transcripts TCV variables

Among the previous studies related to the effects of earnings call transcript tone variables on firm future performance and value uncertainty, (net) positive tone and uncertainty tone variables are especially focused (e.g. Davis et al. 2012; Borochin et al. 2018). Davis et al. (2012) show that net optimistic language in earnings press releases predicts future firm performance, suggesting that managers employ such language to communicate information about expected future firm performance to the market. Meanwhile, market participants regard the managerial language (especially unexpected net optimistic language) in earnings press releases as at least partially credible, despite its potential opportunism. In addition, Borochin et al. (2018) document that net positive tone in conference calls is negatively associated with firm value uncertainty, as reflected in the equity options market. The two studies above reveal that the (net) positive tone variables of earnings call transcripts affect firm asset value and negatively affect asset value volatility and incomplete information, thereby influencing firm credit risk, including bankruptcy risk.

Regarding the economic consequences of uncertainty tones, Kravet and Muslu (2013) document that textual risk disclosures (e.g. uncertainty, risk) in 10-K are linked to higher stock return volatility, and elevate investors' risk perceptions. Owing to the timeliness and interactive nature of earnings call transcripts, the uncertainty tone variables in these transcripts more accurately reflect managers' emotional states under managerial and environmental pressures, thereby having an immediate impact on investors' risk perceptions and firm value risk. Furthermore, greater linguistic uncertainty also heightens external investors' perceptions of incomplete information about the firm's asset value distribution. Thus, we conjecture that the uncertainty tone variables of earnings call transcripts positively affect firm asset value volatility and incomplete information, thereby increasing firm credit risk, including bankruptcy risk.

In addition to the theoretical discussion on the relationship between the aforementioned variables—positive tone and uncertainty tone—and credit (bankruptcy) risk, this study further discusses why these two variables exhibit stronger predictive power compared to other textual feature variables. The possible reasons are as follows:

- (1) The effects of positive tone and uncertainty tone in earnings conference calls on corporate bankruptcy risk are relatively direct and explicit, as these tones influence key theoretical determinants in structural-form credit risk models. Regarding the positive tone of earnings call transcripts, it reflects managerial confidence in the firm's future operations and is closely tied to its going-concern prospects. Consequently, positive tone is positively associated with asset value and, in turn, negatively associated with bankruptcy risk. In addition, although managers may engage in opportunistic behavior when choosing the optimistic language employed in earnings press releases, market participants at least to some extent perceive managerial language as a credible signal (Davis et al. 2012). Specifically, the use of positive tone can alleviate investors' concerns regarding information asymmetry, thereby enhancing market confidence in the firm (Davis et al. 2012; Ning et al. 2025).¹² In contrast, the uncertainty tone of earn-

¹² Ning et al. (2025) show that managerial tone in conference calls conveys credible signals aligned with managers' private information. Analysts interpret these signals accurately, provide confirming feedback, and adjust their forecasts upward (downward) in response to more optimistic (pessimistic) tones.

- ings call transcripts implies that managers may currently be exposed to operational risks arising from managerial or environmental pressures. In particular, uncertainty tone often arises when firms face challenges or lack clear strategies, serving as a direct signal of operational risk. Hence, uncertainty tone is positively associated with asset value volatility and, in turn, with bankruptcy risk. Because these two tone variables directly capture key theoretical determinants in structural-form credit risk models (e.g., incomplete information, asset value volatility, and asset value), they are expected to be particularly effective in predicting bankruptcy risk, compared with other textual features whose influence on these determinants is more ambiguous or indirect.
- (2) Negative tone is typically linked to short-term underperformance or external pressures, reflecting a potential near-term decline in firm asset value while simultaneously easing outside investors' concerns about information incompleteness. Consequently, due to the competing effects of asset value and incomplete information on credit risk, the impact of negative tone on bankruptcy risk remains ambiguous. Furthermore, compared with uncertainty tone, negative tone is less effective in signaling uncontrollable future risks or conveying a lack of managerial confidence.
 - (3) Although modal strong and modal weak tones reflect the intensity of language, their directional implications remain unclear; consequently, their influence on bankruptcy risk is ambiguous in both strength and direction.
 - (4) While litigious tones are associated with legal risk, they do not necessarily indicate bankruptcy risk. In practice, many large firms are able to continue operations even when facing litigation.

Furthermore, tone variables of earnings call transcripts are likely to exhibit greater explanatory power for bankruptcy risk compared to readability measures. This is because readability mainly reflects the ease of text comprehension and structural complexity of text, which is characteristic of pre-prepared and polished documents, rendering it less appropriate for the real-time, interactive nature of earnings calls. In contrast, tone variables—particularly the positive or uncertain tones expressed in Q&A sessions—more directly reveal managerial confidence or concern, thereby more sensitively reflecting a firm's financial pressures and operational risk. Prior studies have also noted that Q&A sections often contain more forward-looking and incremental information than presentation sections (Matsumoto et al. 2011), and that different tones have significant effects on market reactions (Price et al. 2012). Accordingly, tone variables provide a more effective basis for predicting bankruptcy risk than formal readability measures.

Interpretability analyses of *EC_TCV* variables in bankruptcy prediction

To further examine the contribution and directional impact of the selected *EC_TCV* variables on the performance of the bankruptcy prediction model, this study applies Shapley Additive Explanations (SHAP; Lundberg and Lee 2017) to improve model interpretability. SHAP assigns each feature a Shapley value, which represents its marginal contribution to the prediction through an additive attribution framework. In this study, Shapley values are computed for the feature variables in each testing period to assess the next one- to four-year bankruptcy predictions. The corresponding SHAP summary plots (SHAP Feature Importance) are displayed in Figures 1a to 4a (1b to 4b). These plots illustrate the distribution

of Shapley values across all observations. A higher feature value combined with a positive (negative) Shapley value indicates a stronger positive (negative) effect on the model's prediction. In addition, it has to be noted that SHAP Feature Importance is to take the absolute value of the Shapley value of each feature variable. That is, when the absolute value of the Shapley value of the feature variable is larger, its feature importance is stronger.

Figures 1a to 4a (1b to 4b) present the results of the SHAP Summary Plots (SHAP Feature Importance) for the next one- to four-year bankruptcy prediction under the model setting of XGboost model with Easyensemble. The results reveal that (1) Full_POS consistently exhibits relatively high and stable average absolute SHAP values, ranking the top four for the next one- to four-year bankruptcy predictions; (2) Full_UNC ranks within the top six across the same prediction horizons; (3) Full_POS negatively affects bankruptcy risk while Full_UNC has the opposite impact. The first and second findings shed light on why the *EC_TCV* variables substantially improve predictive effectiveness, with their incremental contribution being particularly pronounced in longer-term bankruptcy forecasts. This evidence is consistent with the inherently forward-looking nature of earnings call transcripts. The third finding reveals that earnings call transcripts positive tone variable is negatively related to corporate bankruptcy risk while earnings call transcripts uncertainty tone variable has the opposite effect. Our findings offer insights into the directional impact of *EC_TCV* variables on bankruptcy risk and its term structure.

Figure 1-4. SHAP analyses for the role of earnings call transcripts TCV variables in next one- to four-year bankruptcy prediction

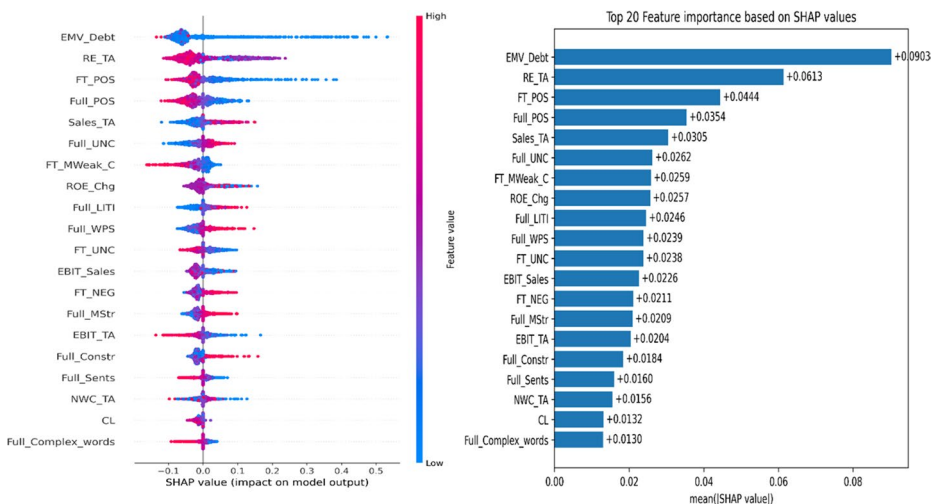


Fig. 1 a. SHAP Summary Plot (Next one-year bankruptcy prediction). **b.** SHAP Feature Importance (Next one-year bankruptcy prediction)

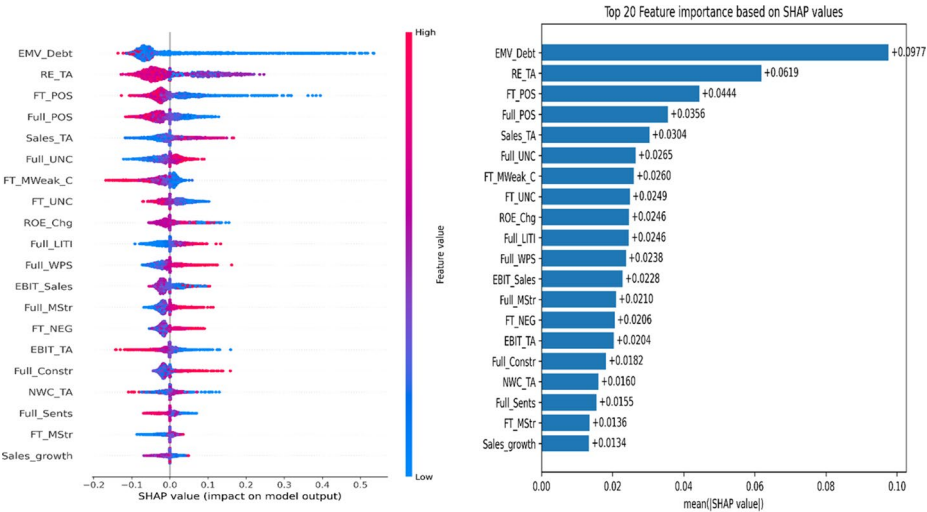


Fig. 2 a. SHAP Summary Plot (Next two-year prediction). b. SHAP Feature Importance (Next two-year bankruptcy prediction)

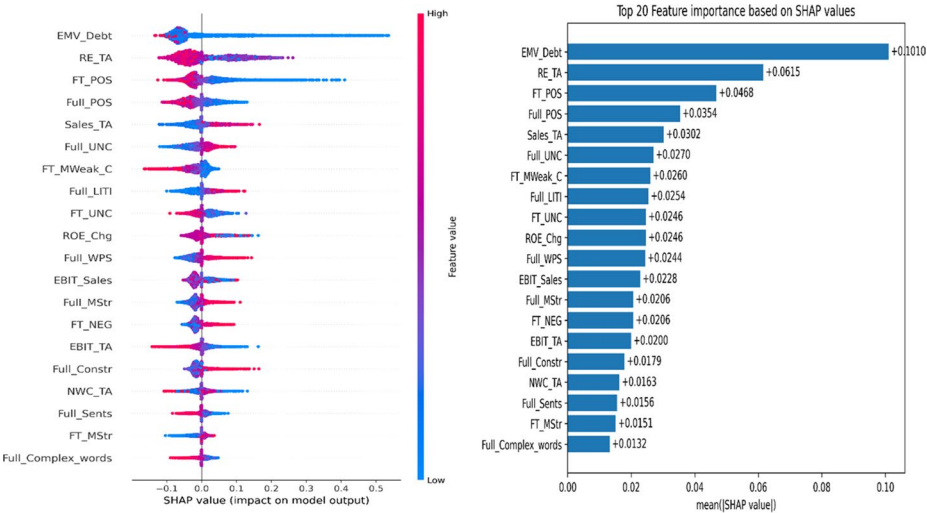


Fig. 3 a. SHAP Summary Plot (Next three-year prediction). b. SHAP Feature Importance (Next three-year bankruptcy prediction)

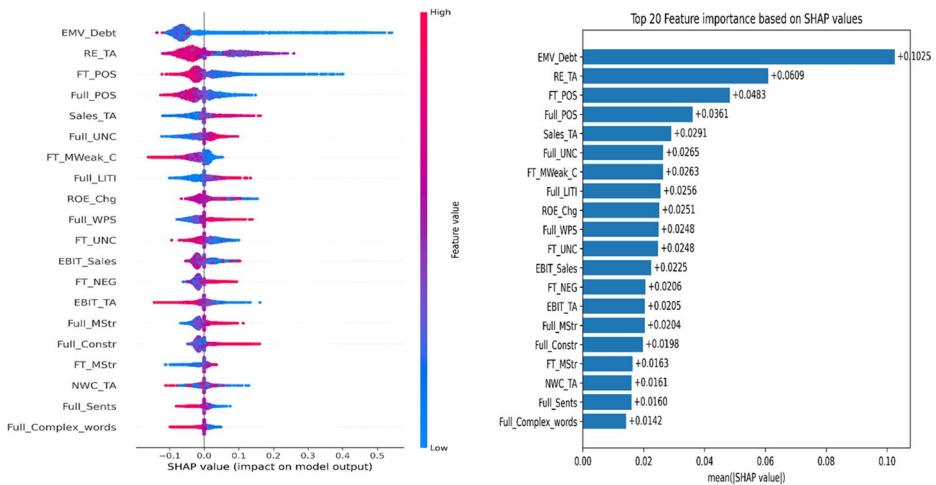


Fig. 4 **a.** SHAP Summary Plot (Next four-year prediction). **b.** SHAP Feature Importance (Next four-year bankruptcy prediction)

Figures 1a to 4a (1b to 4b) present the SHAP summary plots (SHAP Feature Importance) for the next one- to four-year bankruptcy prediction, respectively. The employed features in the SHAP analyses include Barboza et al.'s (2017) financial variables, Chen et al.'s (2023) *TCV* variables, and this study's earnings call transcripts *TCV* (*EC_TCV*) variables. The definitions of these variables are provided in Tables 1 and 2.

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Data availability The datasets used in this study are available on request.

Declarations

Competing interests The authors declare no competing interests.

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